Subject: PROBABILITY & STATISTICS

Subject Code: 23AS4401

UNIT I : Descriptive Statistics

Syllabus: Statistics introduction, population vs sample, collection of data (primary, secondary), measures of central tendency, measures of variability, skewness, kurtosis, correlation, correlation coefficient, rank correlation, regression coefficients, method of least squares, regression lines.

1. Introduction to Statistics

Definition:

Statistics is the science of collecting, organising, presenting, analysing and interpreting numerical data to help in decision making.

- The word comes from the Latin ***status*** / Italian ***statista*** / German ***Statistik*** = "political state".
- Father of Statistics: ***R. A. Fisher***; early work by Sir Francis Galton and Karl Pearson.
- Plural sense: numerical facts (data). Singular sense: the science / methods.

Characteristics of statistics

- Aggregate of facts (a single number is not statistics).
- Numerically expressed and affected by many causes.
- Collected in a systematic manner for a pre-determined purpose.
- Comparable with other data of the same kind.

Functions of statistics

- Simplifies complexity - condenses large data into averages, tables, graphs.
- Presents facts in a definite form; helps in comparison.
- Helps in forecasting, planning and policy making.
- Helps in testing hypotheses and formulating theories.

Limitations

- Deals with groups, not individuals.
- Only quantitative data; results are true on average only.
- Can be misused (wrong sampling, wrong conclusions).

Types of Statistics

Descriptive	Inferential
Summarises data actually collected	Draws conclusions about population from
Mean, median, SD, graphs, correlation	Estimation, hypothesis tests, regression
No generalisation	prediction Generalises with probability

Note:

This course: Unit I = descriptive; Units II-III = probability theory; Units IV-V = inference.

2. Population vs Sample

Definition:

Population: the complete collection of all items / individuals under study.

Sample: a part (subset) of the population selected to represent it.

- Finite population: countable (students in a college). Infinite: uncountable (all possible coin tosses).
- **Parameter**: a numerical measure of a population (μ, σ, P).
- **Statistic**: a numerical measure of a sample ($\bar{x}$, s, p).

Basis	Population	Sample
Size	N (all units)	n (part of N)
Measure called	Parameter	Statistic
Symbols	μ, σ^2, P	$\bar{x}$, s^2, p

Value	Fixed (usually unknown)	Varies sample to sample
Study type	Census / complete enumeration	Sample survey

Census vs Sampling

- Census: every unit is studied - accurate but costly and slow.
- Sampling: faster, cheaper, needed when testing is destructive (bulbs, crash tests).

Methods of sampling (brief)

- ***Simple random***: every unit has equal chance (lottery / random numbers).
- ***Stratified***: divide into homogeneous strata, sample from each.
- ***Systematic***: pick every k-th unit, k = N/n.
- ***Cluster***: choose whole groups (clusters) randomly.
- Non-random: judgement, quota, convenience sampling.

> **Note:**
> Sample must be random and representative, otherwise it gives biased results.

3. Collection of Data

- Data = raw facts / figures. Two sources: primary and secondary.

Primary data

> **Definition:**
> Data collected first-hand by the investigator for the first time, for a specific purpose. It is original.

Methods of collecting primary data

- Direct personal investigation (interview).
- Indirect oral investigation (asking witnesses / third persons).

- Questionnaires: mailed, or filled by enumerators / schedules.
- Local sources / correspondents (newspapers, agencies).
- Observation and experiments.

Features of a good questionnaire

- Short, simple, clear and unambiguous questions.
- Logical order; avoids personal / leading questions.
- Has instructions and a covering letter; pre-tested by a pilot survey.

Secondary data

Definition:
Data already collected by someone else, and available in published or unpublished form.

- Published: government reports (Census, NSSO, RBI bulletin), journals, newspapers, research papers, UN / World Bank reports.
- Unpublished: records of companies, universities, research workers.
- Must be checked for suitability, reliability and adequacy before use.

Primary data	Secondary data
Original, first-hand	Already available, second-hand
More accurate for purpose	May not suit the purpose
Costly and time consuming	Cheap and quick
Needs large manpower	Little manpower
Example: own survey	Example: census report

Classification and presentation

- Raw data is arranged as frequency distribution: discrete (values) or continuous (class intervals).
- Class width h, class mark = (lower + upper) / 2, cumulative frequency (less than / more than).

- Presented by tables, histogram, frequency polygon, ogive, pie and bar diagrams.

4. Measures of Central Tendency

Definition:

An average is a single value that represents the whole data and lies near its centre. A good average is rigidly defined, simple, based on all items, and not affected by extremes.

- Types: Arithmetic mean (AM), Median, Mode, Geometric mean (GM), Harmonic mean (HM).

(a) Arithmetic Mean

Formula:

Ungrouped: $\bar{X} = \Sigma x / n$

Discrete / grouped: $\bar{X} = \Sigma fx / \Sigma f$ (x = value or class mark)

Step deviation: $\bar{X} = A + h \cdot (\Sigma fu / \Sigma f)$, $u = (x - A) / h$

- Properties: $\Sigma(x - \bar{X}) = 0$; $\Sigma(x - \bar{X})^2$ is minimum; affected by extreme values.
- Combined mean of two groups: $\bar{X} = (n1 \cdot \bar{X}1 + n2 \cdot \bar{X}2) / (n1 + n2)$.
- Weighted mean: $\bar{X}w = \Sigma wx / \Sigma w$.

(b) Median (middle value)

- Ungrouped: arrange in order; n odd → value of ((n+1)/2)-th item; n even → average of two middle items.

Formula:

Grouped: $\text{Median} = L + [(N/2 - cf) / f] \times h$

- L = lower limit of median class, cf = cumulative frequency before it, f = its frequency, h = width.
- Not affected by extremes; found graphically by ogive. Quartiles Q1, Q3 and deciles use N/4, 3N/4, ...

(c) Mode (most frequent value)

Formula:

Mode = $L + [(f1 - f0) / (2f1 - f0 - f2)] \times h$

- f1 = frequency of modal class, f0 = before it, f2 = after it.
- A distribution may be unimodal, bimodal or multimodal.

Formula:

Empirical relation: Mode = 3 Median - 2 Mean (moderately skewed data)

(d) Geometric and Harmonic means

Formula:

$GM = (x1 \cdot x2 \cdot \ldots \cdot xn)^{1/n}$; $\log GM = \Sigma \log x / n$

Formula:

$HM = n / \Sigma(1/x)$ [grouped: $\Sigma f / \Sigma(f/x)$]

Note:

For positive unequal values: AM > GM > HM, and $GM^2 = AM \times HM$ (for two values).

- GM used for growth rates, ratios, index numbers; HM used for average speed, rates.

Example:

Find mean, median, mode for the data.

Class: 0-10, 10-20, 20-30, 30-40, 40-50 ; f = 5, 8, 15, 12, 10 (N = 50)

Class	f	x (mark)	fx	cf
0-10	5	5	25	5
10-20	8	15	120	13
20-30	15	25	375	28
30-40	12	35	420	40
40-50	10	45	450	50

- $\Sigma fx = 1390 \rightarrow$ Mean = 1390 / 50 = **27.8**
- N/2 = 25 → median class 20-30 : Median = $20 + (25 - 13)/15 \times 10 = 20 + 8$ = **28**
- Modal class 20-30 : Mode = $20 + (15 - 8)/(30 - 8 - 12) \times 10 = 20 + 7$ = **27**

Note:

Which average to use? Mean for symmetric data, median for skewed/open-ended, mode for most typical value.

5. Measures of Variability (Dispersion)

Definition:

Dispersion measures how much the observations scatter around the average. Two sets can have equal mean but different spread.

- Absolute measures have units; relative measures (coefficients) are pure numbers used to compare.

Types

Measure	Formula	Relative measure
Range	L - S	(L - S) / (L + S)
Quartile deviation	(Q3 - Q1) / 2	(Q3 - Q1) / (Q3 + Q1)
Mean deviation	$\Sigma\|x - A\| / n$	MD / A
Standard deviation	σ	CV = $(\sigma / \bar{x}) \times 100$

Range

Formula:

Range = Largest - Smallest value

- Simplest; depends only on two extreme values.

Mean deviation

Formula:

MD about A = $\Sigma f|x - A| / \Sigma f$ (A = mean or median)

- Least when taken about the median.

Variance and Standard Deviation

Formula:

$\sigma^2 = \Sigma(x - \bar{x})^2 / n = \Sigma x^2/n - (\bar{x})^2$

Grouped: $\sigma^2 = \Sigma f(x - \bar{x})^2 / \Sigma f = \Sigma fx^2/\Sigma f - (\bar{x})^2$

$\sigma = \sqrt{\text{variance}}$; sample variance $s^2 = \Sigma(x - \bar{x})^2 / (n - 1)$

- Properties: independent of change of origin; depends on change of scale: $\sigma(a + bx) = |b|\sigma$.
- Combined SD: $\sigma^2 = [n1(\sigma1^2 + d1^2) + n2(\sigma2^2 + d2^2)] / (n1 + n2)$, $d = \bar{x}i - \bar{x}$.
- Coefficient of variation CV = $(\sigma / \bar{x}) \times 100$ %. Lower CV → more consistent (stable) series.

Example:

Data: 2, 4, 4, 4, 5, 5, 7, 9. Find mean, MD, variance, SD.

Mean = 40/8 = 5. $\Sigma|x - 5| = 3+1+1+1+0+0+2+4 = 12 \rightarrow MD = 12/8 = 1.5$

$\Sigma(x - 5)^2 = 9+1+1+1+0+0+4+16 = 32 \rightarrow \sigma^2 = 4$, $\sigma = 2$, $CV = 2/5 \times 100 = 40\ \%$

Example:

Continue the grouped data of the earlier example (mean = 27.8).

$\Sigma f(x - 27.8)^2 = 7608 \rightarrow \sigma^2 = 7608/50 = 152.16$, $\sigma = 12.34$, $CV = 44.4\ \%$

6. Moments, Skewness and Kurtosis

Moments

Formula:

r-th central moment: $\mu r = \Sigma f\,(x - \bar{x})^r / \Sigma f$

- $\mu 0 = 1$, $\mu 1 = 0$, $\mu 2 = \sigma^2$ (variance).

Formula:

Raw moments about A: $\mu r' = \Sigma f\,(x - A)^r / \Sigma f$

Formula:

$\mu 2 = \mu 2' - (\mu 1')^2$; $\mu 3 = \mu 3' - 3\mu 2'\mu 1' + 2(\mu 1')^3$

$\mu 4 = \mu 4' - 4\mu 3'\mu 1' + 6\mu 2'(\mu 1')^2 - 3(\mu 1')^4$

Skewness

Definition:

Skewness is the lack of symmetry of a distribution.

- Symmetric: mean = median = mode (bell shaped).
- **Positive** skew (tail to right): mean > median > mode.
- **Negative** skew (tail to left): mean < median < mode.

Type	Relation	Sk
Symmetric	$\bar{X} = M = Z$	0
Positively skewed	$\bar{X} > M > Z$	positive
Negatively skewed	$\bar{X} < M < Z$	negative

Measures of skewness

Formula:

Karl Pearson: Sk = (Mean - Mode) / σ or 3(Mean - Median) / σ (limits ±3)

Formula:

Bowley (quartile): Sk = (Q3 + Q1 - 2·Median) / (Q3 - Q1) (limits ±1)

Formula:

Moment measure: $\beta_1 = \mu_3^2 / \mu_2^3$; $\gamma_1 = \sqrt{\beta_1} = \mu_3 / \sigma^3$ (sign of μ_3)

Example:

For the grouped data above: Sk = (27.8 - 27) / 12.34 = 0.065 → almost symmetric, slight positive skew.

Kurtosis

Definition:

Kurtosis measures the peakedness (flatness) of a frequency curve compared with the normal curve.

Formula:

$\beta_2 = \mu_4 / \mu_2^2$; $\gamma_2 = \beta_2 - 3$

Type	β_2	Shape
Mesokurtic	= 3	Normal curve
Leptokurtic	> 3	More peaked, thin tails
Platykurtic	< 3	Flat topped

Note:

For the normal distribution: $\mu_3 = 0$, $\beta_1 = 0$ and $\beta_2 = 3$.

7. Correlation

Definition:

Correlation is the statistical relationship between two variables such that a change in one is accompanied by a change in the other.

- **Positive**: both move in same direction (height - weight).
- **Negative**: move in opposite directions (price - demand).
- Linear / non-linear; simple / multiple / partial.
- Methods: scatter diagram, Karl Pearson's coefficient, Spearman's rank correlation, concurrent deviation.

Scatter diagram

- Points rising upward → positive; falling → negative; no pattern → zero correlation; points on a line → perfect correlation ($r = \pm 1$).

Karl Pearson's coefficient of correlation

Formula:

$r = Cov(x, y) / (\sigma x \cdot \sigma y)$

$Cov(x, y) = \Sigma(x - \bar{x})(y - \bar{y})/n = \Sigma xy/n - \bar{x}\bar{y}$

$r = [n\Sigma xy - (\Sigma x)(\Sigma y)] / \sqrt{\{[n\Sigma x^2 - (\Sigma x)^2] [n\Sigma y^2 - (\Sigma y)^2]\}}$

Properties of r

- $-1 \leq r \leq 1$; r is a pure number (no unit).
- Independent of change of origin and scale (same sign for positive scale).
- Symmetric: $r_{xy} = r_{yx}$.
- r = 0 means no linear relation (not necessarily independence). Independent variables $\Rightarrow$ r = 0.
- Coefficient of determination = r^2; probable error P.E. = $0.6745 (1 - r^2)/\sqrt{n}$.

r value	Interpretation
+1 / -1	Perfect positive / negative
0.7 to 0.99	High degree
0.3 to 0.7	Moderate
0 to 0.3	Low / none

Example:

Find r for x: 1, 2, 3, 4, 5 y: 2, 4, 5, 4, 5

$n = 5, \Sigma x = 15, \Sigma y = 20, \Sigma xy = 66, \Sigma x^2 = 55, \Sigma y^2 = 86$

$r = (5 \times 66 - 15 \times 20) / \sqrt{[(275 - 225)(430 - 400)]} = 30 / \sqrt{1500} =$ **0.775** (high positive)

8. Spearman's Rank Correlation

Definition:

Used when data is given as ranks, or when variables are qualitative (beauty, honesty, ability).

Formula:

$\rho = 1 - 6\Sigma d^2 / [n(n^2 - 1)]$, $d = R1 - R2$

- $-1 \leq \rho \leq 1$. $\rho = 1$ identical ranking; $\rho = -1$ exactly opposite ranking; $\Sigma d = 0$ (check).

Repeated (tied) ranks

- Give each tied item the average of the ranks it occupies (two items at rank 4, 5 each get 4.5).

Formula:

$\rho = 1 - 6\,[\Sigma d^2 + \Sigma(m^3 - m)/12] / [n(n^2 - 1)]$

- m = number of times a value repeats; add one term per repeated group.

Example:

Marks in two tests (9 students): find ρ.

X	35	23	47	17	10	43	9	6	28
Y	30	33	45	23	8	49	12	4	31
Rank	3	5	1	6	7	2	8	9	4
Rank Y	5	3	2	6	8	1	7	9	4
d	-2	2	-1	0	-1	1	1	0	0
d^2	4	4	1	0	1	1	1	0	0

- $\Sigma d^2 = 12$, $n = 9 \rightarrow \rho = 1 - (6 \times 12)/(9 \times 80) = 1 - 0.1 =$ **0.9** (strong agreement)

Note:

Pearson's r uses actual values; Spearman's ρ uses ranks. For no ties ρ equals Pearson's r computed on the ranks.

9. Regression

Definition:

Regression is the statistical method of estimating (predicting) the value of one variable from the known value of another using their average relationship.

- The line of regression of y on x predicts y; of x on y predicts x. Term given by Francis Galton.

Correlation	Regression
Measures degree of relation	Measures nature / form (prediction)
r symmetric in x, y	Regression of y on x ≠ x on y
No cause-effect needed	Dependent and independent variable

Method of least squares

- Fit $y = a + bx$ so that $\Sigma e^2 = \Sigma(y - a - bx)^2$ is minimum.
- Setting partial derivatives to zero gives the ***normal equations***:

Formula:

$$\Sigma y = na + b\Sigma x$$
$$\Sigma xy = a\Sigma x + b\Sigma x^2$$

- Solve for a and b. (Similarly for x on y: $\Sigma x = na + b\Sigma y$, $\Sigma xy = a\Sigma y + b\Sigma y^2$.)

Regression lines and coefficients

Formula:

y on x: $y - \bar{y} = b_{yx}(x - \bar{x})$, $b_{yx} = r\,\sigma y / \sigma x = \mathrm{Cov}(x, y) / \sigma x^2$

Formula:

x on y: $x - \bar{x} = b_{xy}(y - \bar{y})$, $b_{xy} = r\,\sigma x / \sigma y = \mathrm{Cov}(x, y) / \sigma y^2$

Short form: $b_{yx} = [n\Sigma xy - \Sigma x \Sigma y] / [n\Sigma x^2 - (\Sigma x)^2]$

Properties

- Both lines pass through the point $(\bar{x}, \bar{y})$ - their intersection gives the means.
- $r^2 = b_{yx} \cdot b_{xy}$ so $r = \pm\sqrt{(b_{yx}\, b_{xy})}$; r, b_{yx}, b_{xy} have the same sign.
- If one coefficient > 1, the other must be < 1 (product ≤ 1).
- AM of regression coefficients ≥ r.
- Regression coefficients are independent of change of origin but not of scale.
- If $r = \pm 1$ the two lines coincide; if $r = 0$ they are perpendicular ($y = \bar{y}$ and $x = \bar{x}$).

Formula:

Angle between lines: $\tan\theta = [(1 - r^2) / r] \times [\sigma x \sigma y / (\sigma x^2 + \sigma y^2)]$

Example:

For x: 1, 2, 3, 4, 5 and y: 2, 4, 5, 4, 5 find both regression lines.

$\bar{x} = 3$, $\bar{y} = 4$, Cov = 66/5 - 12 = 1.2, $\sigma x^2 = 2$, $\sigma y^2 = 1.2$

$b_{yx} = 1.2/2 = 0.6 \rightarrow y - 4 = 0.6(x - 3) \rightarrow \mathbf{y = 0.6x + 2.2}$

$b_{xy} = 1.2/1.2 = 1 \rightarrow x - 3 = 1(y - 4) \rightarrow \mathbf{x = y - 1}$

Check: $r^2 = 0.6 \times 1 = 0.6 \rightarrow r = 0.775$ ✓. Prediction: $x = 6 \rightarrow y = 5.8$.

Note:

Exam tips: always check ranks sum to $n(n+1)/2$, remember $r = \sqrt{(b_{yx}\, b_{xy})}$ with common sign, and write the formula first.

Quick Revision - Unit I

- Mean = $\Sigma fx/\Sigma f$, Median = $L + (N/2 - cf)h/f$, Mode = $L + (f1-f0)h/(2f1-f0-f2)$.
- $\sigma^2 = \Sigma fx^2/\Sigma f - \bar{x}^2$; $CV = \sigma/\bar{x} \times 100$.
- $Sk = (Mean - Mode)/\sigma$; $\beta 2 = \mu 4/\mu 2^2$ (3 = normal).
- $r = Cov/(\sigma x \sigma y)$; $\rho = 1 - 6\Sigma d^2/n(n^2 - 1)$.
- Regression lines meet at $(\bar{x}, \bar{y})$; $r^2 = b_{yx}\, b_{xy}$.

Subject: PROBABILITY & STATISTICS

Subject Code: 23AS4401

UNIT II : Probability

Syllabus: Probability, probability axioms, addition law and multiplicative law, conditional probability, Bayes' theorem, random variables (discrete and continuous), probability density functions, properties, mathematical expectation.

I. Basic Terms of Probability

Definition:

Random experiment: an experiment whose outcome cannot be predicted in advance, though all possible outcomes are known (tossing a coin, throwing a die).

- ***Sample space (S)***: set of all possible outcomes. Coin: S = {H, T}. Die: S = {1, 2, 3, 4, 5, 6}.
- ***Sample point***: each element of S. ***Event***: any subset of S.
- Simple (elementary) event: one sample point; compound event: more than one.
- ***Sure event*** = S; ***impossible event*** = φ.

Types of events

- ***Mutually exclusive***: cannot occur together, $A \cap B = \varphi$.
- ***Exhaustive***: together cover S, $A \cup B \cup \ldots = S$.
- ***Equally likely***: no outcome is preferred over another.
- ***Independent***: occurrence of one does not affect the other.
- ***Complementary***: A' (or $\bar{A}$) = S – A, "not A".

Set operations as events

Notation	Meaning

$A \cup B$	A or B (at least one) occurs
$A \cap B$	both A and B occur
A'	A does not occur
$A \cap B'$	A occurs but B does not
$A' \cap B'$	neither A nor B occurs

Note:
De Morgan's laws: $(A \cup B)' = A' \cap B'$ and $(A \cap B)' = A' \cup B'$.

2. Definitions of Probability

(a) Classical (Laplace) definition

Formula:
$P(A)$ = (number of favourable cases) / (total number of equally likely cases) = m / n

- Limitations: needs finite and equally likely outcomes.

(b) Statistical (empirical) definition

Formula:
$P(A) = \lim (n \to \infty)$ of m / n (relative frequency in repeated trials)

(c) Axiomatic definition (Kolmogorov, 1933)

Definition:

Let S be a sample space and P a function defined on events. P is a probability if it satisfies the three axioms:

Axiom 1: $P(A) \geq 0$ for every event A (non-negativity)

Axiom 2: $P(S) = 1$ (certainty)

Axiom 3: for mutually exclusive A1, A2, ... : $P(A1 \cup A2 \cup \ldots) = P(A1) + P(A2) + \ldots$

Counting rules (useful for equally likely cases)

Formula:

Permutations: $nPr = n! / (n - r)!$ Combinations: $nCr = n! / [r!(n - r)!]$

- Multiplication rule: if one job can be done in m ways and another in n ways, both can be done in $m \times n$ ways.

Example:

Two dice are thrown. Find P(sum = 7).

Total outcomes = 36. Favourable: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) = 6.

$P = 6/36 = 1/6$

Example:

A committee of 3 is chosen from 5 men and 4 women. P(exactly 2 women)?

$P = (4C2 \times 5C1) / 9C3 = (6 \times 5) / 84 = 30/84 = 5/14$

3. Theorems from the Axioms

- **Theorem 1:** $P(\varphi) = 0$.
  - Proof: $S = S \cup \varphi \cup \varphi \ldots$, all disjoint, so $1 = 1 + P(\varphi) + P(\varphi) + \ldots \Rightarrow P(\varphi) = 0$.

- **Theorem 2 (complement):** $P(A') = 1 - P(A)$.
  - Proof: A and A' are disjoint with $A \cup A' = S \Rightarrow P(A) + P(A') = 1$.
- **Theorem 3:** $0 \leq P(A) \leq 1$.
- **Theorem 4 (monotone):** if $A \subset B$ then $P(A) \leq P(B)$ and $P(B - A) = P(B) - P(A)$.
- **Theorem 5:** $P(A \cap B') = P(A) - P(A \cap B)$.

> **Note:**
> "At least one" problems: P(at least one) = 1 - P(none). It saves a lot of work.

4. Addition Law of Probability

> **Formula:**
> $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

- Proof idea: $A \cup B = A \cup (B \cap A')$, disjoint. $P(B) = P(A \cap B) + P(B \cap A')$. Substitute.
- If A, B are mutually exclusive: $P(A \cup B) = P(A) + P(B)$.

For three events

> **Formula:**
> $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) + P(A \cap B \cap C)$

- Boole's inequality: $P(A \cup B) \leq P(A) + P(B)$.

Example:

A card is drawn from a pack of 52. Find P(king or heart).

P(K) = 4/52, P(H) = 13/52, P(K ∩ H) = 1/52

P = 4/52 + 13/52 - 1/52 = 16/52 = **4/13**

Example:

If P(A) = 0.5, P(B) = 0.4, P(A ∩ B) = 0.2 find P(A ∪ B) and P(neither).

P(A ∪ B) = 0.5 + 0.4 - 0.2 = 0.7 ; P(neither) = 1 - 0.7 = **0.3**

5. Conditional Probability and Multiplication Law

Definition:

The conditional probability of A given that B has occurred is

P(A | B) = P(A ∩ B) / P(B), provided P(B) > 0.

- Similarly P(B | A) = P(A ∩ B) / P(A).
- It reduces the sample space to B.

Multiplication law

Formula:

P(A ∩ B) = P(A) · P(B | A) = P(B) · P(A | B)

P(A ∩ B ∩ C) = P(A) · P(B | A) · P(C | A ∩ B)

Independent events

Definition:

A and B are independent if P(A ∩ B) = P(A) · P(B). (equivalently P(A | B) = P(A))

- If A, B independent, then so are (A, B'), (A', B), (A', B').

- Pairwise independence does not imply mutual independence: for 3 events, need $P(A \cap B \cap C) = P(A)P(B)P(C)$ as well.

Note:

Mutually exclusive $\neq$ independent. If $P(A), P(B) > 0$ and $A \cap B = \varphi$ then they are dependent.

Example:

Urn has 5 red and 3 black balls. Two balls are drawn without replacement. P(both red)?

$P = 5/8 \times 4/7 = 20/56 =$ **5/14** (with replacement it would be $5/8 \times 5/8 = 25/64$)

Example:

Three persons shoot at a target independently with probabilities 1/2, 1/3, 1/4. P(target hit)?

P(none hits) $= 1/2 \times 2/3 \times 3/4 = 1/4 \Rightarrow$ P(at least one hits) $= 1 - 1/4 =$ **3/4**

Example:

If $P(A) = 0.5$, $P(B) = 0.4$, $P(A \cap B) = 0.2$. Are A and B independent?

$P(A)P(B) = 0.2 = P(A \cap B) \Rightarrow$ ***independent***. $P(A \mid B) = 0.2/0.4 = 0.5 = P(A)$.

6. Total Probability and Bayes' Theorem

Definition:

Partition: events B1, B2, ..., Bn are mutually exclusive and exhaustive (B1 ∪ ... ∪ Bn = S) with P(Bi) > 0.

Theorem of total probability

Formula:

$P(A) = \sum P(B_i) \cdot P(A \mid B_i)$ (i = 1 to n)

- Proof: $A = (A \cap B1) \cup ... \cup (A \cap Bn)$, disjoint; apply multiplication law to each term.

Bayes' theorem

Formula:

$P(B_k \mid A) = P(B_k) \cdot P(A \mid B_k) / \sum P(B_i) \cdot P(A \mid B_i)$

- P(Bi) are ***prior*** probabilities; P(Bk | A) is the ***posterior*** probability (revised after A is observed).
- Proof: $P(B_k \mid A) = P(A \cap B_k)/P(A)$; numerator = $P(B_k)P(A \mid B_k)$; denominator by total probability.

Steps to solve Bayes problems

- 1) Name the causes B1, B2, ... and the effect A. 2) Write priors P(Bi).
- 3) Write likelihoods P(A | Bi). 4) Compute P(A) by total probability. 5) Divide.

Example:

Three factories A, B, C make 50 %, 30 %, 20 % of items with defect rates 1 %, 2 %, 3 %. An item is found defective. Find the probability it came from each factory.

Factory	P(Bi)	P(D given Bi)	Product
A	0.5	0.01	0.005

B	0.3	0.02	0.006
C	0.2	0.03	0.006

- P(D) = 0.005 + 0.006 + 0.006 = **0.017**
- P(A | D) = 0.005/0.017 = **5/17** ≈ **0.294**; P(B | D) = 6/17 ≈ 0.353; P(C | D) = 6/17 ≈ 0.353 (sum = 1 ✓)

Example:

A disease affects 1 % of people. A test is positive for 95 % of sick and 5 % of healthy people. A person tests positive. P(sick)?

P(+) = 0.01 × 0.95 + 0.99 × 0.05 = 0.0095 + 0.0495 = 0.059

P(sick | +) = 0.0095 / 0.059 ≈ **0.161** (only 16 %, because the disease is rare)

7. Random Variables

Definition:

A random variable (r.v.) X is a real-valued function defined on the sample space: X : S → R. It assigns a number to every outcome.

- Example: toss 2 coins, X = number of heads; X takes values 0, 1, 2.
- Two types: ***discrete*** (countable values) and ***continuous*** (any value in an interval).

Discrete random variable

- Takes finite or countably infinite values x1, x2, ... (number of defects, number of calls).

Definition:

Probability mass function (p.m.f.): p(xi) = P(X = xi)

Formula:

Properties: (i) $p(x_i) \geq 0$ (ii) $\Sigma p(x_i) = 1$

Formula:

Distribution function (c.d.f.): $F(x) = P(X \leq x) = \Sigma p(x_i)$ over $x_i \leq x$

- F is non-decreasing, step function, right continuous, $F(-\infty) = 0$, $F(\infty) = 1$.
- $P(a < X \leq b) = F(b) - F(a)$; $p(x_i) = F(x_i) - F(x_{i-1})$.

Example:

Two coins tossed, X = number of heads.

X	0	1	2
p(x)	1/4	1/2	1/4
F(x)	1/4	3/4	1

Example:

Let $p(x) = kx$ for $x = 1, 2, 3, 4$. Find k, $P(X \geq 3)$.

$\Sigma p = k(1 + 2 + 3 + 4) = 10k = 1 \Rightarrow k = 0.1$

$P(X \geq 3) = 0.3 + 0.4 = \mathbf{0.7}$

8. Continuous Random Variable and p.d.f.

Definition:

X is continuous if it takes values in an interval and there is a function f(x), the probability density function (p.d.f.), such that

$P(a \leq X \leq b) = \int f(x)\,dx$ from a to b.

Formula:

Properties of p.d.f.: (i) $f(x) \geq 0$ for all x (ii) $\int f(x)\,dx = 1$ over $(-\infty, \infty)$

- For a continuous r.v., $P(X = c) = 0$, hence $P(a \leq X \leq b) = P(a < X < b) = P(a \leq X < b)$ ...
- $f(x)$ is not a probability; it is a density (can exceed 1).

Formula:

c.d.f.: $F(x) = P(X \leq x) = \int f(t)\,dt$ from $-\infty$ to x; $f(x) = dF(x)/dx$

- F is continuous, non-decreasing, $0 \leq F \leq 1$, $F(-\infty) = 0$, $F(+\infty) = 1$.
- $P(a < X < b) = F(b) - F(a)$.
- Median M: $F(M) = 1/2$. Mode: x where $f(x)$ is maximum. Quartiles: $F(Q1) = 1/4$, $F(Q3) = 3/4$.

Example:

Let $f(x) = kx^2$ for $0 \leq x \leq 3$, 0 otherwise. Find k, $P(1 < X < 2)$, $F(x)$.

$\int kx^2\,dx$ from 0 to 3 $= k \times 27/3 = 9k = 1 \Rightarrow$ **$k = 1/9$**

$P(1 < X < 2) = (1/9) \times (8 - 1)/3 =$ **$7/27$**

$F(x) = x^3/27$ for $0 \leq x \leq 3$; 0 for $x < 0$; 1 for $x > 3$.

Example:

A r.v. has $f(x) = 2x$, $0 < x < 1$. Find the median.

$F(M) = M^2 = 1/2 \Rightarrow M = 1/\sqrt{2} \approx$ **0.707**

Example:

Given $f(x) = 3(1 - x)^2$ on $(0, 1)$. Check it is a valid p.d.f.

$\int 3(1 - x)^2\,dx = [-(1 - x)^3]$ from 0 to 1 $= 0 + 1 = 1$ ✓, and $f \geq 0 \Rightarrow$ valid.

9. Mathematical Expectation

Definition:

Expectation (mean) of a r.v. is its long-run average value.

Formula:

Discrete: $E(X) = \Sigma\, x_i \cdot p(x_i)$

Formula:

Continuous: $E(X) = \int x\, f(x)\, dx$ over $(-\infty, \infty)$

Formula:

Function of X: $E[g(X)] = \Sigma\, g(x_i)\, p(x_i)$ or $\int g(x)\, f(x)\, dx$

Properties of expectation

- $E(c) = c$; $E(aX + b) = a\,E(X) + b$.
- $E(X + Y) = E(X) + E(Y)$ (always).
- $E(XY) = E(X)\,E(Y)$ if X, Y are independent.
- If $X \geq 0$ then $E(X) \geq 0$; if $X \leq Y$ then $E(X) \leq E(Y)$.

Variance

Formula:

$Var(X) = E[(X - \mu)^2] = E(X^2) - [E(X)]^2$, $SD = \sqrt{Var}$

- $Var(c) = 0$; $Var(aX + b) = a^2\, Var(X)$.
- $Var(X \pm Y) = Var(X) + Var(Y)$ if X, Y independent.

Moments and M.G.F.

- r-th raw moment $\mu_r' = E(X^r)$; central moment $\mu_r = E[(X - \mu)^r]$.

Formula:

M.G.F.: $M_X(t) = E(e^{tX})$; $E(X^r) = d^rM/dt^r$ at $t = 0$

- $M_{aX+b}(t) = e^{bt} M_X(at)$; for independent X, Y: $M_{X+Y} = M_X \cdot M_Y$.

Example:

Find E(X) and Var(X) for p(x) = 0.1x, x = 1, 2, 3, 4.

$E(X) = 0.1(1 + 4 + 9 + 16) = \mathbf{3}$; $E(X^2) = 0.1(1 + 8 + 27 + 64) = 10$

$Var(X) = 10 - 3^2 = \mathbf{1}$

Example:

For $f(x) = x^2/9$ on (0, 3) find mean and variance.

$E(X) = \int x^3/9\, dx = 81/36 = \mathbf{9/4}$; $E(X^2) = \int x^4/9\, dx = 243/45 = 27/5$

$Var(X) = 27/5 - 81/16 = (432 - 405)/80 = \mathbf{27/80} = 0.3375$

Example:

A game: win Rs 100 with probability 0.2, lose Rs 20 with probability 0.8. Expected gain?

$E = 100(0.2) + (-20)(0.8) = 20 - 16 = \mathbf{Rs\ 4}$ per play (favourable game).

Example:

For $f(x) = 2x$ on (0,1): $E(X) = \int 2x^2\, dx = 2/3$, $E(X^2) = \int 2x^3\, dx = 1/2$, Var $= 1/2 - 4/9 = \mathbf{1/18}$.

10. More Worked Examples

Probability basics

Example:

Three coins are tossed. Find P(at least two heads).

S = {HHH, HHT, HTH, THH, HTT, THT, TTH, TTT}, n(S) = 8.

Favourable: HHH, HHT, HTH, THH = 4 ⇒ P = 4/8 = **1/2**

Example:

Two cards are drawn from a pack without replacement. P(both aces)?

P = 4C2 / 52C2 = 6/1326 = **1/221** (or 4/52 × 3/51)

Example:

P(A) = 1/4, P(B) = 1/3 and P(A ∪ B) = 1/2. Are A, B independent?

P(A ∩ B) = 1/4 + 1/3 – 1/2 = 1/12 = P(A)P(B) ⇒ ***independent***; P(A | B) = (1/12)/(1/3) = 1/4.

Example:

A box has 4 red and 6 blue balls. Two are drawn. P(same colour)?

P = (4C2 + 6C2) / 10C2 = (6 + 15)/45 = **7/15**

Example:

Two students pass independently with probabilities 0.7 and 0.6. Find P(exactly one passes) and P(at least one passes).

P(exactly one) = 0.7 × 0.4 + 0.3 × 0.6 = 0.28 + 0.18 = **0.46**

P(at least one) = 1 – 0.3 × 0.4 = **0.88**

Bayes' theorem

Example:

Bag I has 3 white, 2 black balls; Bag II has 2 white, 4 black. A bag is chosen at random and a ball drawn is white. P(it came from Bag I)?

P(W) = 1/2 × 3/5 + 1/2 × 2/6 = 3/10 + 1/6 = 14/30 = 7/15

P(I | W) = (3/10) / (7/15) = **9/14**

Example:

60 % of students are boys, 40 % girls. 10 % of boys and 5 % of girls are left handed. A left-handed student is chosen. P(girl)?

P(girl | L) = (0.4 × 0.05) / (0.6 × 0.10 + 0.4 × 0.05) = 0.02 / 0.08 = **0.25**

Random variables

Example:

Three coins tossed, X = number of heads. Find the p.m.f., E(X), Var(X).

X	0	1	2	3
p(x)	1/8	3/8	3/8	1/8

- $E(X) = (0 + 3 + 6 + 3)/8 = \mathbf{1.5}$; $E(X^2) = (0 + 3 + 12 + 9)/8 = 3$
- $Var(X) = 3 - 2.25 = \mathbf{0.75}$

Example:

A fair die is thrown. Find E(X) and Var(X).

$E(X) = (1+2+3+4+5+6)/6 = \mathbf{3.5}$; $E(X^2) = 91/6$

$Var(X) = 91/6 - 49/4 = (182 - 147)/12 = \mathbf{35/12} \approx \mathbf{2.92}$

Example:

X takes -1, 0, 1, 2 with probabilities 0.2, 0.3, 0.4, 0.1. Find E(X), Var(X), E(2X + 3), Var(2X + 3).

$E(X) = -0.2 + 0 + 0.4 + 0.2 = \mathbf{0.4}$; $E(X^2) = 0.2 + 0 + 0.4 + 0.4 = 1.0$

$Var(X) = 1 - 0.16 = \mathbf{0.84}$

$E(2X + 3) = 2(0.4) + 3 = \mathbf{3.8}$; $Var(2X + 3) = 4 \times 0.84 = \mathbf{3.36}$

Example:

Sum of two dice: E(sum) = E(X1) + E(X2) = 3.5 + 3.5 = **7** (by linearity of expectation).

Var(sum) = 35/12 + 35/12 = 35/6 (independent dice).

Continuous random variables

Example:

$f(x) = e^{-x}$, $x > 0$. Find $P(X > 2)$, mean, variance, median.

$P(X > 2) = \int e^{-x}\,dx$ from 2 to $\infty = e^{-2} \approx$ **0.1353**

$E(X) = \int x\,e^{-x}\,dx = 1$; $E(X^2) = \int x^2 e^{-x}\,dx = 2 \Rightarrow$ Var = 2 – 1 = **1**

Median: $1 - e^{-M} = 1/2 \Rightarrow M = \ln 2 \approx$ **0.693**

Example:

$f(x) = k(x + 1)$ on (0, 2). Find k, $P(X < 1)$, $E(X)$.

$\int k(x + 1)\,dx = k(2 + 2) = 4k = 1 \Rightarrow$ **k = 1/4**

$P(X < 1) = (1/4)(1/2 + 1) =$ **3/8**

$E(X) = (1/4) \int (x^2 + x)\,dx = (1/4)(8/3 + 2) =$ **7/6**

Example:

M.G.F. of a Bernoulli variable ($P(X=1) = p$, $P(X=0) = q$):

$M(t) = q + p\,e^t \Rightarrow M'(0) = p = E(X)$; $M''(0) = p = E(X^2) \Rightarrow$ Var = $p - p^2 =$ **pq**

II. Practice Problems with Answers

- 1) $P(A) = 0.6$, $P(B) = 0.5$, $P(A \cup B) = 0.8$. Find $P(A \cap B)$ and $P(A \mid B)$.
  - Ans: $P(A \cap B) = 0.3$; $P(A \mid B) = 0.3/0.5 = 0.6$.
- 2) $f(x) = kx(1 - x)$, $0 < x < 1$. Find k, mean, variance.

- Ans: k = 6; mean = 1/2; $E(X^2) = 6(1/4 - 1/5) = 0.3$; variance = 0.3 − 0.25 = 0.05.

- 3) $p(x) = c(2x + 1)$, $x = 0, 1, 2, 3, 4$. Find c, $P(X < 2)$, $E(X)$.
  - Ans: $25c = 1 \Rightarrow c = 1/25$; $P(X < 2) = (1 + 3)/25 = 4/25$; $E(X) = (0+3+10+21+36)/25 = 2.8$.
- 4) A, B independent with $P(A) = 0.3$, $P(B) = 0.4$. Find $P(A \cup B)$, $P(A' \cap B')$.
  - Ans: $0.3 + 0.4 - 0.12 = 0.58$; $P(A' \cap B') = 0.7 \times 0.6 = 0.42$.
- 5) Prove $P(A \cap B') = P(A) - P(A \cap B)$.
  - Hint: $A = (A \cap B) \cup (A \cap B')$, disjoint; apply axiom 3.
- 6) Show $Var(aX + b) = a^2 Var(X)$.
  - Hint: $E(aX + b) = a\mu + b$, so $(aX + b - a\mu - b)^2 = a^2(X - \mu)^2$.

12. Formula Sheet – Unit II

Topic	Formula
Classical	$P(A) = m/n$
Complement	$P(A') = 1 - P(A)$
Addition	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Conditional	$P(A \mid B) = P(A \cap B)/P(B)$
Multiplication	$P(A \cap B) = P(A)P(B \mid A)$
Total prob.	$P(A) = \Sigma P(Bi)P(A \mid Bi)$
Bayes	$P(Bk \mid A) = P(Bk)P(A \mid Bk)/P(A)$
Mean	$E(X) = \Sigma xp$ or $\int xf\,dx$
Variance	$E(X^2) - [E(X)]^2$

Note:
Always draw a tree diagram for Bayes problems – multiply along branches, add across branches.

Quick Revision - Unit II

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$; $P(A \mid B) = P(A \cap B)/P(B)$.
- Independent $\Leftrightarrow P(A \cap B) = P(A)P(B)$.
- Bayes: $P(Bk \mid A) = P(Bk)P(A \mid Bk) / \Sigma P(Bi)P(A \mid Bi)$.
- Discrete: $\Sigma p = 1$; continuous: $\int f = 1$; $F' = f$.
- $E(X) = \Sigma xp$ or $\int xf$; $Var = E(X^2) - [E(X)]^2$.

Subject: PROBABILITY & STATISTICS

Subject Code: 23AS4401

UNIT III : Probability Distributions

Syllabus: Binomial, Poisson and Normal distributions and their properties, Chebyshev's inequality, approximation of the binomial distribution to the normal distribution.

1. Introduction to Probability Distributions

Definition:

A probability distribution describes how the total probability 1 is spread over the values of a random variable.

- Discrete distributions: Bernoulli, Binomial, Poisson, Geometric, Hypergeometric.
- Continuous distributions: Uniform, Exponential, Normal, Gamma, t, χ^2, F.
- This unit: ***Binomial, Poisson, Normal***, Chebyshev's inequality, and normal approximation to binomial.

Distribution	Type	Parameters	Mean	Variance
Binomial	discrete	n, p	np	npq
Poisson	discrete	λ	λ	λ
Normal	continuous	μ, σ^2	μ	σ^2

Note:

Remember: q = 1 – p. Learn this table first – many answers start from it.

2. Bernoulli Trials and Binomial Distribution

Definition:

Bernoulli trial: a trial with only two outcomes - success (probability p) and failure (q = 1 - p).

- Bernoulli r.v.: P(X = 1) = p, P(X = 0) = q; E(X) = p, Var(X) = pq.

Binomial distribution

Definition:

X = number of successes in n independent Bernoulli trials, each with the same probability p of success. We write X ~ B(n, p).

Formula:

$P(X = x) = nCx \cdot p^x \cdot q^{n-x}$, x = 0, 1, 2, ..., n

Conditions

- Fixed number of trials n.
- Each trial has two outcomes (success / failure).
- Trials are independent; p is constant.
- Name: probabilities are the terms of the expansion $(q + p)^n$; so their sum = 1.

Properties of binomial distribution

Formula:

Mean = np, Variance = npq, SD = $\sqrt{(npq)}$

- Mean > Variance (since q < 1).
- Mode: the integer part of (n + 1)p. If (n + 1)p is an integer there are two modes: (n+1)p and (n+1)p - 1.
- Skewness $\gamma 1 = (q - p)/\sqrt{(npq)}$; kurtosis $\gamma 2 = (1 - 6pq)/(npq)$.
- If p = 0.5 symmetric; p < 0.5 positively skewed; p > 0.5 negatively skewed.

- Additive: $X1 \sim B(n1, p)$, $X2 \sim B(n2, p)$ independent $\Rightarrow X1 + X2 \sim B(n1 + n2, p)$.
- Recurrence: $P(x + 1) = [(n - x)/(x + 1)] \cdot (p/q) \cdot P(x)$.

Formula:

M.G.F.: $M(t) = (q + p\,e^{t})^{n}$

Derivation of mean

- $E(X) = \Sigma\, x\, nCx\, p^{x} q^{n-x} = np\, \Sigma\, (n-1)C(x-1)\, p^{x-1} q^{n-x} = np\,(q + p)^{n-1} = np.$

Example:

A fair coin is tossed 6 times. Find P(exactly 4 heads), P(at least 5 heads).

$n = 6, p = 1/2.\ P(X = 4) = 6C4\,(1/2)^{6} = 15/64 \approx$ **0.2344**

$P(X \geq 5) = 6C5/64 + 6C6/64 = (6 + 1)/64 =$ **7/64** $\approx$ **0.1094**

Example:

10 % of bolts are defective. A sample of 5 is taken. Find P(no defective), P(at most 1 defective).

$n = 5, p = 0.1, q = 0.9.\ P(0) = 0.9^{5} = 0.5905$

$P(1) = 5(0.1)(0.9)^{4} = 0.3281 \Rightarrow$ P(at most 1) = 0.5905 + 0.3281 = **0.9186**

Example:

Mean of a binomial is 6 and variance is 4. Find n, p.

$np = 6, npq = 4 \Rightarrow q = 4/6 = 2/3, p = 1/3, n = 6/(1/3) =$ **18**

Example:

If 3 dice are thrown 100 times... for B(n = 10, p = 0.3): mean = 3, variance = 2.1, SD = 1.449.

3. Poisson Distribution

Definition:

Poisson distribution gives the probability of a number of events occurring in a fixed interval of time / space when events occur independently at a constant average rate λ. (Siméon Poisson, 1837)

Formula:

$P(X = x) = e^{-\lambda} \lambda^x / x!$, $x = 0, 1, 2, \ldots$ $X \sim P(\lambda)$

As a limit of binomial

- If $n \to \infty$, $p \to 0$ and $np = \lambda$ stays finite, then $B(n, p) \to \text{Poisson}(\lambda)$.
- Used for rare events: accidents, misprints per page, calls per minute, defects per metre, arrivals at a counter.
- Rule of thumb: $n \geq 20$ and $p \leq 0.05$ (or $n \geq 100$, $np \leq 10$).

Properties

Formula:

Mean = λ, Variance = λ, SD = $\sqrt{\lambda}$

- Mean = Variance is the characteristic property of the Poisson distribution.
- Mode: integer part of λ; if λ is an integer, two modes λ and λ - 1.
- Always positively skewed: $\gamma_1 = 1/\sqrt{\lambda}$, $\gamma_2 = 1/\lambda$.
- Additive: $X_1 \sim P(\lambda_1)$, $X_2 \sim P(\lambda_2)$ independent $\Rightarrow X_1 + X_2 \sim P(\lambda_1 + \lambda_2)$.
- Recurrence: $P(x + 1) = [\lambda/(x + 1)]\, P(x)$, starting $P(0) = e^{-\lambda}$.

Formula:

M.G.F.: $M(t) = \exp[\lambda(e^t - 1)]$

Derivation of mean

- $E(X) = \Sigma\, x\, e^{-\lambda}\lambda^x/x! = \lambda\, e^{-\lambda}\, \Sigma\, \lambda^{x-1}/(x-1)! = \lambda\, e^{-\lambda}\, e^{\lambda} = \lambda$.

Example:

A call centre receives 3 calls per minute on average. Find P(exactly 2 calls) and P(more than 1) in a minute.

$\lambda = 3$, $e^{-3} = 0.0498$. $P(2) = 0.0498 \times 9/2 =$ ***0.2240***

$P(0) = 0.0498$, $P(1) = 0.1494 \Rightarrow P(X > 1) = 1 - 0.1992 =$ ***0.8008***

Example:

A book has 2 misprints per page on average. P(a page has no misprint)?

$\lambda = 2$: $P(0) = e^{-2} =$ ***0.1353***

Example:

1 % of items are defective. In a lot of 200 find P(at most 2 defective) using Poisson.

$\lambda = np = 200 \times 0.01 = 2$. $P(X \leq 2) = e^{-2}(1 + 2 + 2) = 5 \times 0.1353 =$ ***0.6767***

Example:

If $P(X = 1) = P(X = 2)$ for Poisson, find $P(X = 0)$.

$\lambda = \lambda^2/2 \Rightarrow \lambda = 2$. $P(0) = e^{-2} =$ ***0.1353***

Binomial	Poisson
Fixed n, two outcomes	Counts rare events, n large
Parameters n, p	One parameter λ
Mean np > variance npq	Mean = variance = λ
x = 0 to n	x = 0, 1, 2, ... (infinite)

4. Normal Distribution

Definition:

A continuous r.v. X follows the normal distribution with mean μ and standard deviation σ, written $X \sim N(\mu, \sigma^2)$, if its p.d.f. is

Formula:

$$f(x) = [1 / (\sigma\sqrt{(2\pi)})] \cdot \exp[-(x - \mu)^2 / (2\sigma^2)], -\infty < x < \infty$$

- Discovered by De Moivre; developed by Laplace and Gauss ("Gaussian distribution").
- Parameters: $-\infty < \mu < \infty$, $\sigma > 0$. Most important distribution in statistics (heights, marks, errors, ...).

Properties of the normal curve

- Bell shaped, symmetric about $x = \mu$.
- Mean = Median = Mode = μ; Variance = σ^2.
- Skewness = 0; kurtosis $\beta 2 = 3$ (mesokurtic).
- Maximum ordinate at $x = \mu$ equals $1/(\sigma\sqrt{(2\pi)})$.
- Points of inflexion at $x = \mu \pm \sigma$.
- x-axis is an asymptote; total area under curve = 1.
- Mean deviation = 0.7979σ ($\approx 4\sigma/5$); quartile deviation = 0.6745σ; Q.D. : M.D. : S.D. = 10 : 12 : 15.
- Odd order central moments vanish; $\mu 2n = 1 \cdot 3 \cdot 5 \ldots (2n - 1)\ \sigma^{2n}$.

Formula:

M.G.F.: $M(t) = \exp(\mu t + \sigma^2 t^2/2)$

- Linear combinations: $X \sim N(\mu 1, \sigma 1^2)$, $Y \sim N(\mu 2, \sigma 2^2)$ independent $\Rightarrow aX + bY \sim N(a\mu 1 + b\mu 2, a^2\sigma 1^2 + b^2\sigma 2^2)$.

Area property (empirical rule)

Range	Area (probability)
$\mu \pm \sigma$	68.27 %

$\mu \pm 2\sigma$	95.45 %
$\mu \pm 3\sigma$	99.73 %

Standard normal distribution

Definition:

If $X \sim N(\mu, \sigma^2)$, then $Z = (X - \mu)/\sigma \sim N(0, 1)$, with p.d.f. $\varphi(z) = (1/\sqrt{(2\pi)}) \exp(-z^2/2)$.

- Mean 0, variance 1. Table gives $\Phi(z) = P(Z \leq z)$.

Formula:

$P(Z \leq -z) = 1 - \Phi(z)$; $P(a < Z < b) = \Phi(b) - \Phi(a)$; $P(Z > z) = 1 - \Phi(z)$

z	Φ(z)
0.00	0.5000
1.00	0.8413
1.28	0.8997
1.645	0.9500
1.96	0.9750
2.00	0.9772
2.33	0.9901
2.58	0.9951
3.00	0.9987

Note:

Steps: 1) standardise $z = (x - \mu)/\sigma$ 2) sketch the curve 3) use symmetry and the table.

5. Worked Examples on Normal Distribution

Example:

$X \sim N(50, 100)$. Find $P(X > 65)$, $P(40 < X < 60)$.

$\sigma = 10$. $z = (65 - 50)/10 = 1.5 \Rightarrow P(Z > 1.5) = 1 - 0.9332 =$ **0.0668**

$P(40 < X < 60) = P(-1 < Z < 1) = 0.8413 - 0.1587 =$ **0.6826**

Example:

Marks are normal with mean 60, SD 12. What percentage of students score above 75? What is the minimum mark of the top 10 %?

$z = (75 - 60)/12 = 1.25 \Rightarrow P(Z > 1.25) = 1 - 0.8944 = 0.1056 \Rightarrow$ **10.56 %**

Top 10 % $\Rightarrow z = 1.28$. $x = 60 + 1.28 \times 12 =$ **75.4** marks.

Example:

Weights of 1000 students are N(65, 25) kg. How many weigh between 60 and 70 kg?

$\sigma = 5$, $z = \pm 1 \Rightarrow$ area $= 0.6826 \Rightarrow 1000 \times 0.6826 \approx$ **683** students.

Example:

For a normal variate, 7 % of items are below 35 and 89 % are below 63. Find μ and σ.

$P(X < 35) = 0.07 \Rightarrow z = -1.48$; $P(X < 63) = 0.89 \Rightarrow z = 1.23$

$(35 - \mu)/\sigma = -1.48$ and $(63 - \mu)/\sigma = 1.23$. Subtract: $28/\sigma = 2.71 \Rightarrow \sigma = 10.33$, $\mu = 35 + 1.48 \times 10.33 =$ **50.3**

Example:

A machine makes rods of mean length 10 cm and SD 0.05 cm. Rods outside 9.9 – 10.1 are rejected. Find percentage rejected.

$z = \pm 2 \Rightarrow$ accepted 0.9545 $\Rightarrow$ rejected $= 1 - 0.9545 =$ **4.55 %**

6. Chebyshev's Inequality

Definition:

For any random variable X with mean μ and finite variance σ^2, and any k > 0:

Formula:

$P(|X - \mu| \geq k\sigma) \leq 1/k^2$ equivalently $P(|X - \mu| < k\sigma) \geq 1 - 1/k^2$

- Another form: $P(|X - \mu| \geq c) \leq \sigma^2/c^2$ $(c > 0)$.
- It holds for ***every*** distribution, discrete or continuous, so the bound is conservative.

Proof (continuous case)

- $\sigma^2 = \int (x - \mu)^2 f(x)\, dx \geq \int$ over $|x - \mu| \geq k\sigma$ of $(x - \mu)^2 f(x)\, dx$.
- On this region $(x - \mu)^2 \geq k^2\sigma^2$, therefore $\sigma^2 \geq k^2\sigma^2 P(|X - \mu| \geq k\sigma)$.
- Divide by $k^2\sigma^2$: $P(|X - \mu| \geq k\sigma) \leq 1/k^2$. ∎

k	Chebyshev: at least	Normal actual
2	75 %	95.45 %
3	88.9 %	99.73 %
4	93.75 %	99.99 %

Example:

A r.v. has mean 12 and variance 9. Find a lower bound for $P(6 < X < 18)$.

$\sigma = 3$, $|X - 12| < 6 = 2\sigma \Rightarrow k = 2 \Rightarrow P \geq 1 - 1/4 =$ ***0.75***

Example:

For $X \sim B(n = 100, p = 0.5)$: mean 50, $\sigma = 5$. Show $P(40 < X < 60) \geq 0.75$. $k = 2$ gives $P(|X - 50| < 10) \geq 3/4$, i.e. ***at least 0.75***. (Actual value $\approx$ 0.96, far larger.)

Example:

A factory makes items with mean weight 50 g and SD 2 g. Use Chebyshev to bound P(weight outside 44-56).

$|X - 50| \geq 6 = 3\sigma \Rightarrow P \leq 1/9 \approx$ ***0.111***

Note:

Chebyshev is used for the weak law of large numbers: $P(|\bar{X} - \mu| \geq \varepsilon) \leq \sigma^2/(n\varepsilon^2) \to 0$ as $n \to \infty$.

7. Normal Approximation to the Binomial

Definition:

(De Moivre - Laplace theorem) If $X \sim B(n, p)$, then for large n, $Z = (X - np)/\sqrt{(npq)}$ is approximately $N(0, 1)$.

- Thus X is approximately $N(np, npq)$.
- Good approximation when $np > 5$ and $nq > 5$ (or $np \geq 10$, $nq \geq 10$); best when $p \approx 0.5$.
- If p is very small ($np < 5$) use the Poisson approximation instead.

Continuity correction

- Binomial is discrete, normal is continuous: treat integer x as the interval $(x - 0.5, x + 0.5)$.

Binomial	Normal (continuity corrected)

$P(X = a)$	$P(a - 0.5 < X < a + 0.5)$
$P(X \le a)$	$P(X < a + 0.5)$
$P(X \ge a)$	$P(X > a - 0.5)$
$P(a \le X \le b)$	$P(a - 0.5 < X < b + 0.5)$

Example:

A fair coin is tossed 100 times. Find $P(X \ge 60)$ and $P(45 \le X \le 55)$.

$\mu = np = 50$, $\sigma = \sqrt{(100 \times 0.5 \times 0.5)} = 5$.

$P(X \ge 60) = P(X > 59.5)$: $z = (59.5 - 50)/5 = 1.9 \Rightarrow 1 - 0.9713 =$ ***0.0287***

$P(45 \le X \le 55) = P(44.5 < X < 55.5)$: $z = \pm 1.1 \Rightarrow 0.8643 - 0.1357 =$ ***0.7286***

Example:

A die is thrown 180 times. P(number of sixes ≤ 25)?

$n = 180$, $p = 1/6 \Rightarrow \mu = 30$, $\sigma = \sqrt{(180 \times 1/6 \times 5/6)} = \sqrt{25} = 5$.

$P(X \le 25) = P(X < 25.5)$: $z = (25.5 - 30)/5 = -0.9 \Rightarrow$ ***0.1841***

Example:

400 students; each is absent with probability 0.1. P(at most 30 absent)?

$\mu = 40$, $\sigma = \sqrt{(400 \times 0.1 \times 0.9)} = 6$. $P(X < 30.5)$: $z = -9.5/6 = -1.583 \Rightarrow \approx$ ***0.0567***

8. Practice Problems with Answers

- 1) A fair die is thrown 4 times. P(exactly two sixes)? Ans: $4C2\ (1/6)^2(5/6)^2 = 150/1296 = 0.1157$.
- 2) A binomial has mean 4 and variance 3. Find n, p. Ans: $q = 3/4$, $p = 1/4$, $n = 16$.

- 3) Poisson with mean 4: find $P(X \leq 2)$. Ans: $e^{-4}(1 + 4 + 8) = 13 \times 0.0183 = 0.2381$.
- 4) $X \sim N(100, 25)$. Find $P(95 < X < 110)$. Ans: z from -1 to 2: 0.9772 – 0.1587 = 0.8185.
- 5) Show that the normal curve has points of inflexion at $\mu \pm \sigma$. Hint: set $f''(x) = 0$.
- 6) Prove mean of Poisson = variance = λ using the m.g.f. Hint: $M'(0)$, $M''(0)$.

Note:
Quick decision: count with fixed n → binomial; rare events per interval → Poisson; measurements / large n → normal.

9. More on Binomial Distribution

Variance of binomial (derivation)

- $E[X(X - 1)] = \Sigma\, x(x - 1)\, nCx\, p^x q^{n-x} = n(n - 1)p^2 \Sigma\, (n-2)C(x-2)\, p^{x-2} q^{n-x} = n(n - 1)p^2$.
- $Var(X) = E[X(X - 1)] + E(X) - [E(X)]^2 = n(n - 1)p^2 + np - n^2p^2 = np - np^2 = \mathbf{npq}$.

Fitting a binomial distribution

- Step 1: find p from the data (mean = np). Step 2: compute P(x). Step 3: expected frequency = N × P(x).

Example:
4 coins are tossed 160 times. Find expected frequencies of 0, 1, 2, 3, 4 heads.

Heads x	0	1	2	3	4
P(x)	1/16	4/16	6/16	4/16	1/16
Expected f	10	40	60	40	10

Example:

Ten fair coins tossed. Find $P(X = 5)$ and $P(X \leq 2)$.

$P(5) = 10C5/1024 = 252/1024 = \mathbf{0.2461}$

$P(X \leq 2) = (1 + 10 + 45)/1024 = 56/1024 = \mathbf{0.0547}$

Example:

The chance a seed germinates is 0.25. 8 seeds are sown. Find P(at least one germinates).

$P = 1 - 0.75^8 = 1 - 0.1001 = \mathbf{0.8999}$

Example:

Find the mode of $B(n = 10, p = 0.4)$.

$(n + 1)p = 11 \times 0.4 = 4.4 \Rightarrow$ mode $= \mathbf{4}$

Example:

A multiple choice test has 5 questions, each with 4 options. A student guesses. P(at least 4 correct)?

$p = 1/4$. $P(4) = 5\,(1/4)^4(3/4) = 15/1024$; $P(5) = 1/1024$.

$P(X \geq 4) = 16/1024 = \mathbf{0.0156}$

Example:

In a binomial with $n = 5$, find the variance of the number of heads when 5 coins are tossed.

$npq = 5 \times 0.5 \times 0.5 = \mathbf{1.25}$

Note:

Maximum variance of $B(n, p)$ for fixed n occurs at $p = 1/2$ (npq is largest when $pq = 1/4$).

10. More on Poisson Distribution

Variance of Poisson (derivation)

- $E[X(X-1)] = \Sigma\, x(x-1)\, e^{-\lambda}\lambda^x/x! = \lambda^2 e^{-\lambda} \Sigma\, \lambda^{x-2}/(x-2)! = \lambda^2$.
- $\text{Var}(X) = \lambda^2 + \lambda - \lambda^2 = \lambda$.

Fitting a Poisson distribution

- Mean of data $\bar{x} = \Sigma fx/\Sigma f = \lambda$. Expected frequency $= N \cdot e^{-\lambda}\lambda^x/x!$, use recurrence $P(x+1) = \lambda P(x)/(x+1)$.

Example:

Accidents occur at an average of 1.5 per day. Find P(0), P(1), P(2) and P(at least 3 accidents).

x	0	1	2
P(x)	0.2231	0.3347	0.2510

- $P(X \le 2) = 0.2231 + 0.3347 + 0.2510 = 0.8088 \Rightarrow P(X \ge 3) = 1 - 0.8088 =$ **0.1912**

Example:

A binomial with n = 100, p = 0.02: use Poisson to find P(X = 3).

$\lambda = np = 2$. $P(3) = e^{-2} \times 2^3/3! = 0.1353 \times 1.3333 =$ **0.1804**

Example:

2 % of fuses are defective. In a box of 100 find P(at least one defective).

$\lambda = 2 \Rightarrow P(X \ge 1) = 1 - e^{-2} = 1 - 0.1353 =$ **0.8647**

Example:

A Poisson variable has $P(X = 0) = 0.3679$. Find its mean and $P(X = 2)$.

$e^{-\lambda} = 0.3679 \Rightarrow \lambda = 1$. $P(2) = e^{-1}/2 =$ **0.1839**

Example:

Number of emails per hour is Poisson with mean 4. Find P(at most 1 email in a half hour).

Half hour: $\lambda = 2$. $P(X \le 1) = e^{-2}(1 + 2) = 3 \times 0.1353 =$ **0.4060**

Note:

If the average rate is λ per unit time, then in t units the mean becomes λt.

II. More on Normal Distribution

Total area under the curve = 1 (proof sketch)

- Let $I = \int \exp(-z^2/2)\, dz$. Then $I^2 = \iint \exp(-(x^2+y^2)/2)\, dx\, dy = \int$ from 0 to $2\pi \int$ from 0 to $\infty \exp(-r^2/2)\, r\, dr\, d\theta = 2\pi$.
- So $I = \sqrt{(2\pi)}$ and $\int \varphi(z)\, dz = 1$.

Mean and variance of $N(\mu, \sigma^2)$

- Put $z = (x - \mu)/\sigma$: $E(X) = \mu + \sigma E(Z)$, and $E(Z) = \int z\, \varphi(z)\, dz = 0$ (odd function) $\Rightarrow E(X) = \mu$.
- $Var(Z) = \int z^2\, \varphi(z)\, dz = 1$ (integration by parts) $\Rightarrow Var(X) = \sigma^2$.

Using the table – special areas

Required	Formula
$P(Z \le a)$, $a > 0$	$\Phi(a)$ directly
$P(Z \ge a)$, $a > 0$	$1 - \Phi(a)$
$P(-a \le Z \le a)$	$2\Phi(a) - 1$
$P(Z \le -a)$	$1 - \Phi(a)$

$P(a \le Z \le b)$	$\Phi(b) - \Phi(a)$

Example:

$X \sim N(30, 16)$. Find $P(X < 24)$ and $P(26 < X < 38)$.

$\sigma = 4$. $z = (24 - 30)/4 = -1.5 \Rightarrow P =$ **0.0668**

$P(26 < X < 38) = P(-1 < Z < 2) = 0.9772 - 0.1587 =$ **0.8185**

Example:

$X \sim N(70, 64)$. Find k with $P(X > k) = 0.05$.

$z = 1.645 \Rightarrow k = 70 + 1.645 \times 8 =$ **83.16**

Example:

Scores are $N(80, 100)$. Find percentage between 70 and 90, and above 100.

Between: $z = \pm 1 \Rightarrow$ **68.26 %**. Above 100: $z = 2 \Rightarrow 1 - 0.9772 =$ **2.28 %**

Example:

$X \sim N(10, 4)$ and $Y \sim N(20, 9)$ independent. Find the distribution of $X + Y$ and $X - Y$, and $P(X + Y > 35)$.

$X + Y \sim N(30, 13)$; $X - Y \sim N(-10, 13)$ (variances always add).

$z = (35 - 30)/\sqrt{13} = 5/3.606 = 1.39 \Rightarrow P = 1 - 0.9177 =$ **0.0823**

Example:

Heights of 500 men are $N(168, 36)$ cm. How many are taller than 180 cm?

$z = 12/6 = 2 \Rightarrow 0.0228 \times 500 \approx$ **11** men.

Example:

The life of a bulb is N(1200, 3600) hours. Find the life L such that 90 % of bulbs last more than L.

$P(X > L) = 0.9 \Rightarrow z = -1.28 \Rightarrow L = 1200 - 1.28 \times 60 =$ **1123.2 hours**

12. More on Chebyshev and Approximations

Example:

Mean 100, SD 10. At least what fraction lies between 80 and 120?

$k = 2 \Rightarrow$ at least $1 - 1/4 =$ **75 %**

Example:

How many standard deviations around the mean guarantee at least 96 % of the data?

$1 - 1/k^2 = 0.96 \Rightarrow k^2 = 25 \Rightarrow$ **$k = 5$**

Example:

The number of items sold has mean 80, variance 25. Lower bound for $P(70 < X < 90)$?

$|X - 80| < 10 = 2\sigma \Rightarrow P \geq$ **0.75**

Example:

Using Chebyshev, estimate the number of tosses so that $P(|\hat{p} - 0.5| < 0.1) \geq 0.95$.

$\sigma^2 = 0.25/n$; need $0.25/(n \times 0.01) \leq 0.05 \Rightarrow n \geq$ **500**

Normal approximation – more examples

Example:

A fair coin is tossed 200 times. Find $P(X \geq 110)$.

$\mu = 100$, $\sigma = \sqrt{50} = 7.07$. $P(X > 109.5)$: $z = 9.5/7.07 = 1.34 \Rightarrow 1 - 0.9099 = \approx$ **0.090**

Example:

$X \sim B(20, 0.5)$. Approximate $P(X = 10)$.

$\mu = 10$, $\sigma = 2.236$. $P(9.5 < X < 10.5) = P(-0.22 < Z < 0.22) = 2(0.5871) - 1 =$ **0.174** (exact 0.1762)

Example:

Binomial $n = 500$, $p = 0.004$ – which approximation? $np = 2 < 5 \Rightarrow$ **Poisson with $\lambda = 2$**, not normal.

Distribution	Approximated by	When
Binomial	Poisson(np)	n large, p small
Binomial	Normal(np, npq)	n large, p not small
Poisson	Normal(λ, λ)	λ large (≥ 10)

13. Practice Problems with Answers

- 1) A coin is tossed 5 times. P(exactly 3 heads)? Ans: 10/32 = 0.3125.
- 2) A binomial has $n = 6$ and $9P(X = 4) = P(X = 2)$. Find p. Ans: $9 \times 15\, p^4 q^2 = 15\, p^2 q^4 \Rightarrow 9p^2 = q^2 \Rightarrow 3p = q \Rightarrow p = 1/4$.
- 3) Poisson mean 3: $P(X > 4)$? Ans: $1 - e^{-3}(1 + 3 + 4.5 + 4.5 + 3.375) = 1 - 0.8153 = 0.1847$.
- 4) $X \sim N(50, 16)$. Find $P(X > 58)$. Ans: $z = 2 \Rightarrow 0.0228$.
- 5) Prove that $E(X) = \lambda$ for Poisson using the m.g.f. Hint: $M'(t) = \lambda e^t M(t)$.

- 6) Show $\beta_2 = 3$ for the normal distribution. Hint: $\mu_4 = 3\sigma^4$.
- 7) A r.v. has mean 5 and variance 4. Bound $P(|X - 5| \geq 6)$. Ans: $\leq 4/36 = 1/9$.
- 8) 30 % of items are defective; sample of 100: approximate $P(X \leq 35)$. Ans: $\mu = 30$, $\sigma = 4.58$, $z = 5.5/4.58 = 1.2 \Rightarrow 0.8849$.

Quick Revision – Unit III

- Binomial: $nCx\ p^x q^{n-x}$; mean np, var npq.
- Poisson: $e^{-\lambda}\lambda^x/x!$; mean = var = λ.
- Normal: $Z = (X - \mu)/\sigma$; 68 – 95 – 99.7 rule.
- Chebyshev: $P(|X - \mu| < k\sigma) \geq 1 - 1/k^2$.
- Binomial → Normal: $N(np, npq)$ with continuity correction ± 0.5.

Subject: PROBABILITY & STATISTICS

Subject Code: 23AS4401

UNIT IV : Estimation and Testing of Hypothesis, Large Sample Tes

Syllabus: Estimation – parameters, statistics, sampling distribution, point estimation, null and alternative hypothesis, critical and acceptance regions, level of significance, two types of errors, power of test; large sample tests for proportions and means; confidence intervals.

I. Parameters, Statistics and Sampling Distribution

Definition:

Statistical inference: drawing conclusions about a population from a sample. It has two branches – ***estimation*** and ***testing of hypothesis***.

- ***Parameter***: numerical characteristic of the population (μ, σ, P) – fixed, usually unknown.
- ***Statistic***: function of sample observations ($\bar{x}$, s, p) – a random variable.
- ***Estimator***: statistic used to estimate a parameter; ***estimate***: its numerical value for a given sample.

Definition:

Sampling distribution: the probability distribution of a statistic computed from all possible samples of size n drawn from the population.

Standard error (S.E.): the standard deviation of the sampling distribution of a statistic.

Sampling distributions (large n)

Statistic	Mean	Standard error
Sample mean $\bar{x}$	μ	$\sigma/\sqrt{n}$
Sample proportion p	P	$\sqrt{(PQ/n)}$

Difference of means $\bar{x}1 - \bar{x}2$	$\mu1 - \mu2$	$\sqrt{(\sigma1^2/n1 + \sigma2^2/n2)}$
Difference of proportions $p1 - p2$	$P1 - P2$	$\sqrt{(P1Q1/n1 + P2Q2/n2)}$

- Finite population (sampling without replacement): multiply S.E. by $\sqrt{[(N - n)/(N - 1)]}$.

Central Limit Theorem

Definition:

If $x1, \ldots, xn$ is a random sample from any population with mean μ and variance σ^2, then for large n ($n \geq 30$) the sample mean $\bar{X}$ is approximately $N(\mu, \sigma^2/n)$.

- Hence $Z = (\bar{X} - \mu)/(\sigma/\sqrt{n}) \sim N(0, 1)$. This is why normal tests work for large samples.
- If the population itself is normal, $\bar{X}$ is exactly normal for every n.

Note:

Large sample: $n \geq 30$. Small sample: $n < 30$ (Unit V). If σ is unknown, use the sample s in place of σ for large n.

2. Point Estimation

Definition:

Point estimation gives a single value (a point) as the estimate of an unknown parameter, e.g. $\bar{X}$ for μ.

Properties of a good estimator

- **Unbiasedness:** $E(T) = \theta$. Example: $E(\bar{X}) = \mu$; $E(s^2) = \sigma^2$ with $s^2 = \Sigma(x - \bar{X})^2/(n - 1)$.
  - The estimator with divisor n has $E = [(n - 1)/n]\,\sigma^2$ and is biased.

- **Consistency:** T converges to θ as n → ∞ (i.e. E(T) → θ and Var(T) → 0).
- **Efficiency:** among unbiased estimators the one with the smallest variance is most efficient. Efficiency = Var(T2)/Var(T1).
  - Example: for a normal population $\bar{x}$ is more efficient than the sample median (Var(median) ≈ $\pi\sigma^2/2n$).
- **Sufficiency:** T contains all the information about θ in the sample (Fisher - Neyman factorisation).

Methods of point estimation

- **Method of moments:** equate population moments to sample moments.
- **Maximum likelihood (MLE):** choose θ that maximises L(θ) = f(x1, θ) f(x2, θ) ... f(xn, θ). Solve d log L/dθ = 0.

Population	MLE
Binomial B(n, p)	$\hat{p} = x/n$
Poisson(λ)	$\hat{\lambda} = \bar{x}$
Normal N(μ, σ^2)	$\hat{\mu} = \bar{x}$, $\hat{\sigma}^2 = \Sigma(x - \bar{x})^2/n$
Exponential(θ)	$\hat{\theta} = 1/\bar{x}$

Example:

Find MLE of λ for Poisson.

$L = \Pi e^{-\lambda}\lambda^{xi}/xi! \Rightarrow \log L = -n\lambda + (\Sigma xi)\log\lambda - \Sigma\log xi!$

$d\log L/d\lambda = -n + \Sigma xi/\lambda = 0 \Rightarrow \hat{\lambda} = \Sigma xi/n = \bar{x}$

Example:

Sample 4, 6, 8, 10, 12 from a population. Give point estimates of μ and σ^2.

$\bar{x} = 40/5 = 8$. $s^2 = (16 + 4 + 0 + 4 + 16)/4 = 10$.

3. Confidence Interval Estimation

Definition:

A $(1-\alpha) \times 100$ % confidence interval is a random interval that contains the true parameter with probability $1-\alpha$.

- 95 % CI means: if we repeat sampling many times, about 95 % of such intervals contain the parameter.

Confidence level	α	Critical z (two-sided)
90 %	0.10	1.645
95 %	0.05	1.96
98 %	0.02	2.33
99 %	0.01	2.58

One-sample problems

Formula:

Mean (σ known or n large): $\bar{x} \pm z_{\alpha/2} \cdot \sigma/\sqrt{n}$ (use s if σ unknown)

Formula:

Proportion: $p \pm z_{\alpha/2} \cdot \sqrt{(pq/n)}$

Two-sample problems

Formula:

Difference of means: $(\bar{x}1 - \bar{x}2) \pm z_{\alpha/2} \sqrt{(\sigma 1^2/n1 + \sigma 2^2/n2)}$

Formula:

Difference of proportions: $(p1 - p2) \pm z_{\alpha/2} \sqrt{(p1q1/n1 + p2q2/n2)}$

- Margin of error $E = z_{\alpha/2} \times$ S.E. Wider interval $\Leftrightarrow$ higher confidence or smaller n.

Formula:

Sample size for mean: $n = (z_{\alpha/2}\ \sigma / E)^2$ for proportion: $n = z^2\ pq / E^2$

Example:

A sample of 100 has mean 52 and SD 10. Find 95 % and 99 % confidence limits for μ.

S.E. = $10/\sqrt{100} = 1$. 95 %: 52 ± 1.96 = **(50.04, 53.96)**

99 %: 52 ± 2.58 = **(49.42, 54.58)**

Example:

In a sample of 500, 40 % favour a scheme. Find 95 % CI for the population proportion.

$p = 0.4$, S.E. = $\sqrt{(0.4 \times 0.6/500)} = 0.0219$. Limits: $0.4 \pm 1.96 \times 0.0219 = 0.4 \pm 0.0429$ = **(0.357, 0.443)**

Example:

Samples: $n1 = 50$, $\bar{X}1 = 82$, $s1 = 8$; $n2 = 60$, $\bar{X}2 = 78$, $s2 = 6$. Find 95 % CI for $\mu1 - \mu2$.

S.E. = $\sqrt{(64/50 + 36/60)} = \sqrt{1.88} = 1.371$. $4 \pm 1.96 \times 1.371 = 4 \pm 2.69$ = **(1.31, 6.69)**

Interval does not contain 0, so the means differ significantly at 5 %.

Example:

How large a sample is needed to estimate a mean within 3 units with 95 % confidence if $\sigma = 15$?

$n = (1.96 \times 15/3)^2 = (9.8)^2 = 96.04 \Rightarrow$ **$n = 97$**

Example:

Sample sizes n1 = 400, n2 = 500; p1 = 0.15, p2 = 0.12. Find 95 % CI for P1 – P2.

S.E. = $\sqrt{(0.15 \times 0.85/400 + 0.12 \times 0.88/500)} = \sqrt{0.00053} = 0.0230$

$0.03 \pm 1.96 \times 0.0230 =$ **(-0.015, 0.075)** - contains 0 ⇒ no significant difference.

4. Testing of Hypothesis – Basic Concepts

Definition:

Statistical hypothesis: a statement / assumption about a population parameter or distribution, which may or may not be true.

Definition:

Testing of hypothesis: a rule that decides, on the basis of sample data, whether to accept or reject the hypothesis.

Null and alternative hypotheses

- **Null hypothesis H0:** the hypothesis to be tested, stated as "no difference" or "no effect" (e.g. H0: $\mu = \mu_0$).
- **Alternative hypothesis H1 (Ha):** what we accept if H0 is rejected.
  - H1: $\mu \neq \mu_0$ → **two-tailed** test.
  - H1: $\mu > \mu_0$ (right-tailed) or $\mu < \mu_0$ (left-tailed) → **one-tailed** test.
- Simple hypothesis fixes all parameters; composite does not.

Test statistic, critical and acceptance regions

- **Test statistic:** a function of the sample whose distribution under H0 is known (e.g. Z).

- **Critical region (rejection region):** set of values of the test statistic for which H0 is rejected.
- **Acceptance region:** the remaining values where H0 is not rejected.
- Critical value: the boundary between the two regions.

Level of significance

Definition:

Level of significance α = P(rejecting H0 when H0 is true). Commonly 5 % or 1 %.

Level α	Two-tailed z	Right-tailed z	Left-tailed z
10 %	1.645	1.28	-1.28
5 %	1.96	1.645	-1.645
1 %	2.58	2.33	-2.33

Two types of errors

Decision	H0 true	H0 false
Reject H0	**Type I error** (α)	Correct (power)
Accept H0	Correct	**Type II error** (β)

- **Type I error:** rejecting H0 when it is true; P = α (producer's risk).
- **Type II error:** accepting H0 when it is false; P = β (consumer's risk).
- Reducing α increases β for a fixed sample size; both fall if n increases.

Power of a test

Formula:

Power = $1 - \beta$ = P(rejecting H0 when H1 is true)

- A good test has small α and large power. Most powerful test: highest power for a given α.

Example:

H0: $\mu = 100$ against H1: $\mu = 105$, $\sigma = 10$, $n = 25$, $\alpha = 0.05$ (right-tailed). Find β and power.

S.E. $= 10/5 = 2$. Critical value: $\bar{x} = 100 + 1.645 \times 2 = 103.29$. Reject H0 if $\bar{x} > 103.29$.

$\beta = P(\bar{x} < 103.29 \mid \mu = 105) = P(z < -0.855) =$ **0.196**; power = **0.804**

5. General Procedure for Testing

- Step 1: State H0 and H1 (decide one- or two-tailed).
- Step 2: Choose level of significance α.
- Step 3: Choose the test statistic and compute its value from the sample.
- Step 4: Find the critical value from the table.
- Step 5: Compare: if $|z_{cal}| > z_{tab}$ (in the critical region) reject H0; otherwise accept (do not reject) H0.
- Step 6: Write the conclusion in words.

Note:

Accepting H0 only means there is no sufficient evidence against it – it does not prove H0 is true.

Which test to use?

Situation	Test statistic
Single mean, n large	$z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$
Two means	$z = (\bar{x}_1 - \bar{x}_2)/S.E.$
Single proportion	$z = (p - P_0)/\sqrt{(P_0Q_0/n)}$
Two proportions	$z = (p_1 - p_2)/S.E.$ (pooled)
Small samples	t, F or χ^2 tests (Unit V)

6. Large Sample Tests – Single Mean

Formula:

H0: $\mu = \mu_0$. $z = (\bar{x} - \mu_0) / (\sigma/\sqrt{n})$ [use s if σ unknown] ~ N(0, 1)

- Conditions: $n \geq 30$ (or population normal with σ known).

Example:

A sample of 100 items has mean 52 and the population SD is 10. Test H0: $\mu = 50$ at 5 % level.

H0: $\mu = 50$, H1: $\mu \neq 50$ (two-tailed), $\alpha = 0.05$, $z_{tab} = 1.96$.

$z = (52 - 50)/(10/10) = 2$. $|z| = 2 > 1.96 \Rightarrow$ ***reject H0*** at 5 %.

At 1 % ($z_{tab} = 2.58$): $2 < 2.58 \Rightarrow$ accept H0.

Example:

A company claims bulbs last 1600 hours. A sample of 64 bulbs has mean 1570 hours and s = 120. Test at 5 % whether the claim is false (left-tailed).

H0: $\mu = 1600$, H1: $\mu < 1600$. $z = (1570 - 1600)/(120/8) = -30/15 = -2$.

$z_{tab} = -1.645$. Since $-2 < -1.645 \Rightarrow$ ***reject H0***; mean life is less than claimed.

Example:

A sample of 64 has mean 48.5 and s = 4. Test H0: $\mu = 50$ at 1 %.

$z = (48.5 - 50)/(4/8) = -3$. $|z| = 3 > 2.58 \Rightarrow$ ***reject H0***.

Example:

Average marks of 36 students is 70 with σ = 12. Is it consistent with population mean 75 at 5 %?

$z = (70 - 75)/(12/6) = -2.5$, $|z| > 1.96 \Rightarrow$ ***not consistent*** (reject H0).

7. Large Sample Tests – Difference of Means

Formula:

H0: $\mu_1 = \mu_2$. $z = (\bar{x}_1 - \bar{x}_2) / \sqrt{(\sigma_1^2/n_1 + \sigma_2^2/n_2)}$

If $\sigma_1 = \sigma_2 = \sigma$ (common): $z = (\bar{x}_1 - \bar{x}_2) / [\sigma \sqrt{(1/n_1 + 1/n_2)}]$; if σ unknown replace σ^2 by s^2.

Example:

Two groups: $n_1 = 50$, $\bar{x}_1 = 82$, $s_1 = 8$ and $n_2 = 60$, $\bar{x}_2 = 78$, $s_2 = 6$. Is there a significant difference at 5 % and 1 %?

H0: $\mu_1 = \mu_2$, H1: $\mu_1 \neq \mu_2$. S.E. $= \sqrt{(64/50 + 36/60)} = \sqrt{1.88} = 1.371$.

$z = 4/1.371 = 2.92$. $2.92 > 1.96$ and $2.92 > 2.58 \Rightarrow$ ***reject H0*** at both levels – significant difference.

Example:

Mean income in city A ($n = 400$) is Rs 54 with SD 8; in city B ($n = 600$) Rs 52 with SD 6. Is A richer? (5 %, right-tailed)

S.E. $= \sqrt{(64/400 + 36/600)} = \sqrt{(0.16 + 0.06)} = 0.469$. $z = 2/0.469 = 4.26 > 1.645 \Rightarrow$ ***reject H0***: A has higher mean.

Example:

Two machines: n1 = n2 = 100, means 12.1 and 11.9, both SD = 1. Test equality at 5 %.

S.E. = $\sqrt{(1/100 + 1/100)} = 0.1414$. $Z = 0.2/0.1414 = 1.414 < 1.96 \Rightarrow$ ***accept H0***.

Note:

Steps never change - only the S.E. changes with the test.

8. Large Sample Tests - Single Proportion

Formula:

H0: $P = P0$. $Z = (p - P0) / \sqrt{(P0\ Q0 / n)}$, $p = x/n$, $Q0 = 1 - P0$

- Conditions: $nP0 > 5$ and $nQ0 > 5$ (large n).

Example:

In a sample of 400 items, 52 are defective. Test the claim that 10 % are defective at 5 %.

$p = 52/400 = 0.13$. S.E. $= \sqrt{(0.1 \times 0.9/400)} = 0.015$.

$Z = (0.13 - 0.10)/0.015 = 2 > 1.96 \Rightarrow$ ***reject H0***: claim not supported.

Example:

A coin is tossed 400 times and shows 216 heads. Is it fair? (5 %)

$p = 0.54$, $P0 = 0.5$, S.E. $= \sqrt{(0.25/400)} = 0.025$. $Z = 0.04/0.025 = 1.6 < 1.96$ $\Rightarrow$ ***accept H0*** (fair).

Example:

A manufacturer claims at least 95 % of goods are non-defective. Of 600 items, 570 are good. Test at 5 % (left-tailed).

$p = 570/600 = 0.95$. $Z = 0 \Rightarrow$ ***accept H0*** – claim holds.

Example:

In a survey of 1000 voters, 540 favour a candidate. Test H0: $P = 0.5$ against $P > 0.5$ at 1 %.

$p = 0.54$, S.E. $= \sqrt{(0.25/1000)} = 0.0158$. $Z = 0.04/0.0158 = 2.53 > 2.33 \Rightarrow$ ***reject H0***.

9. Large Sample Tests – Difference of Proportions

Formula:

H0: $P1 = P2$. $Z = (p1 - p2) / \sqrt{[\hat{P}\hat{Q}(1/n1 + 1/n2)]}$

Pooled proportion: $\hat{P} = (x1 + x2)/(n1 + n2) = (n1p1 + n2p2)/(n1 + n2)$, $\hat{Q} = 1 - \hat{P}$

Example:

In city A 60 of 400 smoke; in city B 60 of 500 smoke. Is the difference significant at 5 %?

$p1 = 0.15$, $p2 = 0.12$. $\hat{P} = 120/900 = 0.1333$, $\hat{Q} = 0.8667$.

S.E. $= \sqrt{(0.1333 \times 0.8667 \times (1/400 + 1/500))} = \sqrt{(0.11556 \times 0.0045)} = 0.0228$

$Z = 0.03/0.0228 = 1.32 < 1.96 \Rightarrow$ ***accept H0***: no significant difference.

Example:

Before a campaign 500 of 2000 favoured a product; after it 600 of 2000. Has the campaign been effective? (5 %, one-tailed)

$p1 = 0.25$, $p2 = 0.30$. $\hat{P} = 1100/4000 = 0.275$, $\hat{Q} = 0.725$. S.E. $= \sqrt{(0.275 \times 0.725 \times 2/2000)} = 0.01412$

$Z = (0.30 - 0.25)/0.01412 = 3.54 > 1.645 \Rightarrow$ ***reject H0***: campaign effective.

Example:

200 of 1000 tablets of type A and 170 of 1000 of type B were found defective. Test equality.

$\hat{P} = 370/2000 = 0.185$. S.E. $= \sqrt{(0.185 \times 0.815 \times 0.002)} = 0.01736$. $Z = 0.03/0.01736 = 1.73 < 1.96 \Rightarrow$ ***accept H0***.

10. Practice Problems with Answers

- 1) A sample of 49 gives mean 24 and SD 7. 95 % CI for μ? Ans: S.E. = 1, CI = (22.04, 25.96).
- 2) A sample proportion is 0.2 with n = 400. 99 % CI for P? Ans: S.E. = 0.02; $0.2 \pm 0.0516 = (0.148, 0.252)$.
- 3) Test H0: $\mu = 60$ if $n = 100$, $\bar{X} = 62$, $s = 8$ at 5 %. Ans: $Z = 2.5 > 1.96$, reject H0.
- 4) 45 of 100 patients recover with drug A, 60 of 120 with drug B. Test P1 = P2 at 5 %. Ans: $\hat{P} = 105/220 = 0.477$, S.E. = 0.0674, $Z = 0.05/0.0674 = 0.74$, accept.
- 5) Two samples: $n1 = 100$, $\bar{X}1 = 25$, $s1 = 3$; $n2 = 150$, $\bar{X}2 = 24$, $s2 = 4$. Test equality at 5 %. Ans: S.E. $= \sqrt{(0.09 + 0.1067)} = 0.4435$, $Z = 2.25$, reject.
- 6) Explain Type I and Type II errors with an example. Hint: court trial – convicting an innocent person is Type I; acquitting a guilty person is Type II.

- 7) Define level of significance and power of a test.

> **Note:**
> Writing tips: always state H0, H1, α, test statistic, calculated value, tabulated value, and the final conclusion in words.

II. More Worked Examples – Sampling and Estimation

Sampling distribution of the mean (illustration)

> **Example:**
> Population: 2, 4, 6 ($\mu = 4$, $\sigma^2 = 8/3$). All samples of size 2 with replacement: find the sampling distribution of $\bar{X}$.

Sample	(2,2)	(2,4)	(2,6)	(4,2)	(4,4)	(4,6)	(6,2)	(6,4)	(6,6)
$\bar{X}$	2	3	4	3	4	5	4	5	6

- Mean of all $\bar{X} = 36/9 = 4 = \mu$ ($\bar{X}$ is unbiased).
- Variance of $\bar{X} = (4 + 1 + 1 + 0 + 0 + 0 + 1 + 1 + 4)/9 = 12/9 = 4/3 = \sigma^2/n = (8/3)/2$ ✓.

Why $\bar{X}$ is unbiased and consistent

- $E(\bar{X}) = (1/n)\,\Sigma\, E(x_i) = n\mu/n = \mu$. $Var(\bar{X}) = n\sigma^2/n^2 = \sigma^2/n \to 0$ as $n \to \infty$.
- Therefore $\bar{X}$ is an unbiased and consistent estimator of μ.

Example:

A sample of 36 has mean 70 and σ = 12. Find 95 % and 90 % confidence limits for μ.

S.E. = 12/6 = 2. 95 %: 70 ± 3.92 = **(66.08, 73.92)**

90 %: 70 ± 1.645 × 2 = 70 ± 3.29 = **(66.71, 73.29)** (lower confidence ⇒ narrower interval)

Example:

Of 600 men, 300 and of 400 women, 325 prefer a brand. Give a test of equal preference.

$p_1 = 0.5$, $p_2 = 0.8125$, $\hat{P} = 625/1000 = 0.625$. S.E. = $\sqrt{(0.625 \times 0.375 \times (1/600 + 1/400))} = 0.03125$

$Z = 0.3125/0.03125 = 10 > 2.58 \Rightarrow$ ***reject H0*** - women's preference is much higher.

12. P-value and One-Tailed vs Two-Tailed Tests

Definition:

P-value: the probability, assuming H0 is true, of getting a test statistic at least as extreme as the one observed.

- Rule: if p-value < α reject H0; if p-value ≥ α accept H0.
- Two-tailed: $p = 2 \times P(Z > |z|)$. One-tailed: $p = P(Z > z)$ or $P(Z < z)$.

Example:

$Z = 2.0$ (two-tailed): $p = 2(1 - 0.9772) =$ ***0.0456*** $< 0.05 \Rightarrow$ reject at 5 % but not at 1 %.

Example:

$Z = 1.6$ (two-tailed): $p = 2(1 - 0.9452) =$ **0.1096** $> 0.05 \Rightarrow$ accept H0.

Choosing the tail

Wording in the problem	H1
"is different", "changed", "not equal"	two-tailed ($\neq$)
"greater", "higher", "more than", "improved"	right-tailed ($>$)
"less", "lower", "fewer", "reduced"	left-tailed ($<$)

- Critical region for two-tailed test is split: $\alpha/2$ in each tail. For one-tailed all α is in one tail.

Note:

One-tailed tests are more powerful in the stated direction but cannot detect change in the other direction.

13. Errors and Power – More Examples

Example:

To test H0: $p = 0.5$ against H1: $p = 0.75$ a coin is tossed 5 times. Reject H0 if 4 or more heads occur. Find α, β and power.

$\alpha = P(X \geq 4 \mid p = 0.5) = (5 + 1)/32 =$ **0.1875**

$\beta = P(X \leq 3 \mid p = 0.75) = 1 - [5(0.75)^4(0.25) + 0.75^5] = 1 - (0.3955 + 0.2373) =$ **0.3672**

Power $= 1 - \beta =$ **0.6328**

- Effect of changing things (other things fixed):
  - Increase n $\Rightarrow$ power increases, both α and β can be reduced.
  - Increase $\alpha \Rightarrow \beta$ decreases $\Rightarrow$ power increases.

- Larger true difference $|\mu_1 - \mu_0| \Rightarrow$ power increases.

- Neyman – Pearson lemma: for simple H0 vs simple H1, the most powerful test of size α rejects H0 when the likelihood ratio $L(\theta_1)/L(\theta_0)$ is large.

14. More Large Sample Tests – Examples

Example:

A die is thrown 600 times and a six appears 120 times. Is the die unbiased? (5 % and 1 %)

H0: P = 1/6. p = 0.2. S.E. = $\sqrt{(1/6 \times 5/6 / 600)} = 0.01521$. $Z = 0.0333/0.01521 = 2.19$.

$2.19 > 1.96 \Rightarrow$ reject at 5 %; $2.19 < 2.58 \Rightarrow$ accept at 1 %. ***Conclusion depends on the level.***

Example:

Heights of 400 men: mean 171.38 cm, population mean claimed 171.17 cm, σ = 3.3 cm. Test at 5 %.

$Z = (171.38 - 171.17)/(3.3/20) = 0.21/0.165 = 1.27 < 1.96 \Rightarrow$ ***accept H0.***

Example:

Two brands of tyres. Brand 1: n = 100, mean 1200 hours, s = 36. Brand 2: n = 100, mean 1180 hours, s = 40. Is brand 1 better? (5 %, one-tailed)

S.E. = $\sqrt{(36^2/100 + 40^2/100)} = \sqrt{(12.96 + 16)} = 5.38$. $Z = 20/5.38 = 3.72 > 1.645 \Rightarrow$ ***reject H0.***

Example:

A sample of 100 bolts has mean length 3.02 cm; the process mean is 3.00 cm with $\sigma = 0.1$ cm. Is the machine in order? (5 %)

$Z = 0.02/(0.1/10) = 2 > 1.96 \Rightarrow$ ***reject H0*** – machine out of adjustment.

Example:

20 % of a lot is claimed defective. A sample of 100 has 14 defective. Test H0: $P = 0.2$ against $P < 0.2$ at 5 %.

S.E. $= \sqrt{(0.2 \times 0.8/100)} = 0.04$. $Z = (0.14 - 0.2)/0.04 = -1.5 > -1.645 \Rightarrow$ ***accept H0***.

Note:

Compare $|Z|$ with the table value; for left-tailed tests compare the negative values carefully.

15. More Practice Problems

- 8) Sample of 200 gives mean 15 and SD 3. Find 98 % confidence limits. Ans: S.E. = 0.212, $15 \pm 2.33 \times 0.212 = (14.51, 15.49)$.
- 9) Can a sample of 100 with mean 49 and $s = 5$ come from a population with $\mu = 50$? (5 %) Ans: $Z = -2$, $|Z| > 1.96$, no.
- 10) 15 out of 60 boys and 25 out of 90 girls are vegetarian. Test equality of proportions at 5 %. Ans: $\hat{P} = 40/150 = 0.267$, S.E. $= \sqrt{(0.267 \times 0.733 \times (1/60 + 1/90))} = 0.0739$, $Z = (0.25 - 0.278)/0.0739 = -0.38$, accept.
- 11) Find the sample size to estimate a proportion within 0.05 at 95 % confidence when p is unknown (use $pq = 0.25$). Ans: $n = 1.96^2 \times 0.25/0.0025 = 384.16 \Rightarrow 385$.
- 12) Distinguish between parameter and statistic; estimate and estimator.

- 13) State the Central Limit Theorem and explain its use in large sample tests.

Quick Revision – Unit IV

- S.E. of $\bar{x} = \sigma/\sqrt{n}$; of $p = \sqrt{(PQ/n)}$.
- CI: estimate $\pm$ z $\times$ S.E. (z = 1.96 for 95 %, 2.58 for 99 %).
- α = P(Type I); β = P(Type II); power = 1 – β.
- Critical z (two-tailed): 1.645 (10 %), 1.96 (5 %), 2.58 (1 %).
- Reject H0 if $|z_{cal}| > z_{tab}$.
- Pooled $\hat{P} = (x1 + x2)/(n1 + n2)$ for the difference of proportions.

Subject: PROBABILITY & STATISTICS

Subject Code: 23AS4401

UNIT V : Small Sample Tests

Syllabus: Student t-distribution (single mean, two means, paired t-test), F-test for equality of variances, chi-square test for goodness of fit and for independence of attributes.

1. Small Samples and Exact Sampling Distributions

Definition:

A sample is small if $n < 30$. The normal (z) test is not reliable because s is a poor estimate of σ; exact sampling distributions – t, χ^2 and F – are used.

- Assumption: the parent population is (approximately) normal.
- **Degrees of freedom (d.f. = ν):** number of independent observations free to vary = n minus the number of constraints (parameters estimated from the sample).

Test	Distribution	Used for
t-test	Student's t	Means (single, two, paired)
F-test	Snedecor's F	Equality of two variances
Chi-square	χ^2	Goodness of fit, independence

Note:

Same six steps as Unit IV: H0, H1, α, test statistic, table value, conclusion.

2. Student's t-distribution

Definition:

If x1, ..., xn is a random sample from $N(\mu, \sigma^2)$, then $t = (\bar{x} - \mu) / (s/\sqrt{n})$ follows Student's t-distribution with $\nu = n - 1$ degrees of freedom. (W. S. Gosset, "Student", 1908.)

Formula:

p.d.f.: $f(t) = C_\nu (1 + t^2/\nu)^{-(\nu+1)/2}$, $-\infty < t < \infty$

- Also $t = z / \sqrt{(\chi^2/\nu)}$, the ratio of a standard normal to the square root of an independent χ^2 divided by its d.f.

Properties

- Bell shaped and symmetric about $t = 0$ (like normal) but with heavier tails.
- Mean $= 0$ $(\nu > 1)$; Variance $= \nu/(\nu - 2)$ $(\nu > 2) > 1$.
- As $\nu \to \infty$, t-distribution tends to $N(0, 1)$; for $\nu \geq 30$ the normal is a good approximation.
- Shape depends only on ν (one parameter). Table gives $t_\alpha(\nu)$, the point with upper tail area α.
- Odd moments are zero.

d.f. ν	5 % two-tailed	1 % two-tailed	5 % one-tailed
5	2.571	4.032	2.015
7	2.365	3.499	1.895
8	2.306	3.355	1.860
9	2.262	3.250	1.833
10	2.228	3.169	1.812
15	2.131	2.947	1.753
20	2.086	2.845	1.725
24	2.064	2.797	1.711
30	2.042	2.750	1.697
∞	1.960	2.576	1.645

Applications of the t-distribution

- Test for single mean; test for difference of two means (independent samples); paired t-test.
- Test of significance of correlation coefficient; confidence intervals for the mean of a small sample.

3. t-test for a Single Mean

Formula:

H0: $\mu = \mu_0$. $t = (\bar{x} - \mu_0) / (s/\sqrt{n})$, $s^2 = \Sigma(x - \bar{x})^2/(n - 1)$, d.f. $= n - 1$

- If the S.D. is computed with divisor n (call it S), use $t = (\bar{x} - \mu_0)/(S/\sqrt{(n - 1)})$.
- Reject H0 if $|t_{cal}| > t_{tab}(n - 1, \alpha)$.

Confidence limits for μ

Formula:

$\bar{x} \pm t_{\alpha/2}(n - 1) \cdot s/\sqrt{n}$

Example:

Ten values: 50, 52, 48, 51, 49, 53, 47, 50, 52, 48. Test H0: $\mu = 48$ against $\mu \neq 48$ at 5 %.

$\bar{x} = 500/10 = 50$. Deviations: 0, 2, −2, 1, −1, 3, −3, 0, 2, −2; $\Sigma(x - \bar{x})^2 = 36$.

$s^2 = 36/9 = 4 \Rightarrow s = 2$. $t = (50 - 48)/(2/\sqrt{10}) = 2/0.6325 =$ ***3.16***

$t_{tab}(9) = 2.262$. $3.16 > 2.262 \Rightarrow$ ***reject H0***: mean differs from 48.

Example:

For the same data find 95 % confidence limits for μ.

$50 \pm 2.262 \times 0.6325 = 50 \pm 1.43 =$ **(48.57, 51.43)** (48 is outside, consistent with rejection).

Example:

A sample of 25 has mean 102 and s = 10. Test H0: μ = 100 at 5 %.

$t = (102 - 100)/(10/5) = 1$. $t_{tab}(24) = 2.064$. $1 < 2.064 \Rightarrow$ ***accept H0***.

Example:

A machine is to fill 500 g packets. 16 packets give mean 497 g, s = 6 g. Is the machine under-filling? (5 %, left-tailed)

$t = (497 - 500)/(6/4) = -2$. $t_{tab}(15)$ one-tailed 5 % $= -1.753$. $-2 < -1.753 \Rightarrow$ ***reject H0***: under-filling.

Example:

Nine students' marks: mean 56, s = 6 (divisor n – 1). Test μ = 60 at 5 %.

$t = (56 - 60)/(6/3) = -2$. $t_{tab}(8) = 2.306$. $|t| < 2.306 \Rightarrow$ ***accept H0***.

4. t-test for Difference of Two Means (independent samples)

- Assumptions: both populations normal; independent samples; equal variances $\sigma1^2 = \sigma2^2$ (check with F-test first).

Formula:

H0: $\mu_1 = \mu_2$. $t = (\bar{x}_1 - \bar{x}_2) / [\, s\sqrt{(1/n_1 + 1/n_2)}\,]$, d.f. $= n_1 + n_2 - 2$

Pooled variance: $s^2 = [\Sigma(x_1 - \bar{x}_1)^2 + \Sigma(x_2 - \bar{x}_2)^2] / (n_1 + n_2 - 2) = [(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2] / (n_1 + n_2 - 2)$

Example:

Sample A: 12, 15, 18, 14, 16. Sample B: 10, 13, 11, 12, 14. Is there a significant difference between means at 5 %?

$\bar{x}_1 = 75/5 = 15$, $\Sigma(x - \bar{x}_1)^2 = 9 + 0 + 9 + 1 + 1 = 20$.

$\bar{x}_2 = 60/5 = 12$, $\Sigma(x - \bar{x}_2)^2 = 4 + 1 + 1 + 0 + 4 = 10$.

$s^2 = (20 + 10)/8 = 3.75 \Rightarrow s = 1.936$. S.E. $= 1.936 \times \sqrt{(2/5)} = 1.2247$

$t = 3/1.2247 =$ ***2.45***. $t_{tab}(8) = 2.306 \Rightarrow 2.45 > 2.306 \Rightarrow$ ***reject H0*** at 5 %.

At 1 % $t_{tab}(8) = 3.355 > 2.45 \Rightarrow$ accept H0 at 1 %.

Example:

Two batches of 10 and 12 items have means 20 and 18 with $s_1 = 3$, $s_2 = 4$. Test equality at 5 %.

$s^2 = (9 \times 9 + 11 \times 16)/20 = (81 + 176)/20 = 12.85$, $s = 3.585$.

S.E. $= 3.585\sqrt{(1/10 + 1/12)} = 3.585 \times 0.4282 = 1.535$. $t = 2/1.535 = 1.30 < t_{tab}(20) = 2.086 \Rightarrow$ ***accept H0***.

Note:

If variances are unequal (F-test rejects) use Welch's approximate t-test – beyond this syllabus.

5. Paired t-test

Definition:

Used when observations are paired (same subjects before and after, matched pairs). The two samples are dependent and have equal size n.

- Let $d_i = x_i - y_i$ (differences). Test whether the mean difference is zero.

Formula:

H0: $\mu d = 0$. $t = \bar{d} / (s_d/\sqrt{n})$, $s_d^2 = \Sigma(d - \bar{d})^2/(n - 1)$, d.f. = n – 1

- It reduces variation between subjects and is more powerful than the independent test when pairs are correlated.

Example:

Marks of 8 students before and after coaching. Test if coaching is effective at 5 % (one-tailed).

Student	1	2	3	4	5	6	7	8
Before	70	72	68	75	80	66	74	71
After	74	75	73	77	86	70	77	76
After – Before	4	3	5	2	6	4	3	5

- $\Sigma d = 32 \Rightarrow \bar{d} = 4$; $\Sigma(d - \bar{d})^2 = 0 + 1 + 1 + 4 + 4 + 0 + 1 + 1 = 12$.
- $s_d^2 = 12/7 = 1.714 \Rightarrow s_d = 1.309$. $t = 4/(1.309/\sqrt{8}) = 4/0.4629 =$ ***8.64***
- $t_{tab}(7)$, one-tailed 5 % = 1.895. $8.64 > 1.895 \Rightarrow$ ***reject H0***: coaching is effective.

Example:

Blood pressure of 5 patients before and after a drug: differences d = 6, 4, 8, 2, 5. Test H0: $\mu d = 0$ (5 %, two-tailed).

$\bar{d} = 5$. $\Sigma(d - \bar{d})^2 = 1 + 1 + 9 + 9 + 0 = 20 \Rightarrow s_d^2 = 5$, $s_d = 2.236$.

$t = 5/(2.236/\sqrt{5}) = 5/1 = 5$. $t_{tab}(4) = 2.776 \Rightarrow 5 > 2.776 \Rightarrow$ ***reject H0***.

6. F-distribution and Test for Equality of Variances

Definition:

If $s1^2$ and $s2^2$ are variances of independent samples from normal populations, then $F = s1^2/s2^2$ (when $\sigma1^2 = \sigma2^2$) follows Snedecor's F-distribution with $(\nu1, \nu2) = (n1 - 1, n2 - 1)$ d.f.

Formula:

$F = (\chi1^2/\nu1) / (\chi2^2/\nu2)$ (ratio of two independent chi-squares each divided by its d.f.)

Properties of the F-distribution

- Positively skewed, defined for $F > 0$; shape depends on $(\nu1, \nu2)$.
- Mean $= \nu2/(\nu2 - 2)$ for $\nu2 > 2$.
- Reciprocal property: $F_{1-\alpha}(\nu1, \nu2) = 1 / F_{\alpha}(\nu2, \nu1)$.
- If t has ν d.f. then t^2 has $F(1, \nu)$ distribution.

F-test procedure

Formula:

$H0: \sigma1^2 = \sigma2^2$. F = (larger variance) / (smaller variance) $= s_L^2 / s_S^2 \geq 1$

- d.f. $= (n_L - 1, n_S - 1)$, where n_L is the size of the sample with the larger variance.
- Reject H0 if $F_{cal} > F_{tab}(\nu1, \nu2)$ at the chosen α (for a two-tailed test use $\alpha/2$ table value).

Some F values (5 %, upper)	$(\nu1, \nu2)$
3.29	(7, 9)
3.35	(10, 8)
2.98	(10, 10)

5.05	(5, 5)
6.39	(4, 4)

Example:

Sample 1: $n = 8$, $\Sigma(x - \bar{x})^2 = 84$. Sample 2: $n = 10$, $\Sigma(x - \bar{x})^2 = 45$. Test equality of variances at 5 % (one-tailed).

$s1^2 = 84/7 = 12$; $s2^2 = 45/9 = 5$. $F = 12/5 = \mathbf{2.4}$, d.f. (7, 9).

$F_{tab}(7, 9) = 3.29$. $2.4 < 3.29 \Rightarrow$ ***accept H0***: variances may be taken equal.

Example:

$n1 = 11$, $s1^2 = 25$; $n2 = 9$, $s2^2 = 16$. Test $\sigma1^2 = \sigma2^2$ at 5 %.

$F = 25/16 = 1.5625$, d.f. (10, 8), $F_{tab} = 3.35 \Rightarrow$ ***accept H0***.

Note:

Do the F-test first; if variances are equal, then use the pooled t-test for two means.

7. Chi-square Distribution

Definition:

If $Z1, \ldots, Z\nu$ are independent standard normal variables then $\chi^2 = Z1^2 + \ldots + Z\nu^2$ follows chi-square distribution with ν d.f.

Formula:

p.d.f.: $f(\chi^2) = C \cdot e^{-\chi^2/2} (\chi^2)^{\nu/2 - 1}$, $\chi^2 > 0$

Properties

- Mean = ν; Variance = 2ν; Mode = $\nu - 2$ ($\nu \geq 2$).

- Positively skewed; tends to normal $N(\nu, 2\nu)$ for large ν. χ^2 is never negative.
- Additive: χ^2 with $\nu1$ d.f. plus independent χ^2 with $\nu2$ d.f. gives χ^2 with $\nu1 + \nu2$ d.f.
- For a normal sample: $(n - 1)s^2/\sigma^2 \sim \chi^2(n - 1)$ (basis for variance tests).

d.f.	5 % point	1 % point
1	3.841	6.635
2	5.991	9.210
3	7.815	11.345
4	9.488	13.277
5	11.070	15.086
6	12.592	16.812
8	15.507	20.090
10	18.307	23.209

Conditions for applying χ^2 tests

- Total frequency N is large (at least 50) and observations are independent.
- Each expected frequency ≥ 5; if smaller, pool neighbouring classes (and reduce d.f. accordingly).
- Frequencies are actual counts, not percentages.

8. χ^2 Test for Goodness of Fit

Definition:

Tests whether observed frequencies agree with those expected under a theoretical distribution (uniform, binomial, Poisson, normal, 9:3:3:1 ratio, …).

Formula:

H0: the data fit the distribution. $\chi^2 = \Sigma (O - E)^2 / E$

d.f. = k – 1 – m (k = number of classes, m = number of parameters estimated from the data)

- Reject H0 if $\chi^2_{cal} > \chi^2_{tab}(d.f., \alpha)$. It is always a right-tailed test.

Example:

A die is thrown 120 times with outcomes 1 to 6 observed as 15, 25, 20, 18, 22, 20. Is it fair (5 %)?

Face	1	2	3	4	5	6
O	15	25	20	18	22	20
E	20	20	20	20	20	20
$(O - E)^2/E$	1.25	1.25	0	0.2	0.2	0

- $\chi^2 = 1.25 + 1.25 + 0 + 0.2 + 0.2 + 0 = \mathbf{2.9}$; d.f. = 5; $\chi^2_{tab} = 11.07 \Rightarrow 2.9 < 11.07 \Rightarrow$ ***accept H0***: die is fair.

Example:

Mendel: peas in the ratio 9:3:3:1 expected. Observed 315, 108, 101, 32 (N = 556). Test at 5 %.

Class	O	E (9:3:3:1)	$(O - E)^2/E$
Round yellow	315	312.75	0.0162
Round green	108	104.25	0.1349
Wrinkled yellow	101	104.25	0.1013
Wrinkled green	32	34.75	0.2176

- $\chi^2 = 0.0162 + 0.1349 + 0.1013 + 0.2176 = \mathbf{0.470}$; d.f. = 3; $\chi^2_{tab} = 7.815 \Rightarrow$ ***accept H0***; data fit the 9:3:3:1 ratio very well.

Example:

Fitting a binomial: d.f. = k – 1 – 1 = k – 2 if p is estimated from data; k – 1 if p is given.

Example:

200 families with 4 children each; observed boys 0 to 4: 12, 52, 76, 44, 16. Do they fit B(4, 1/2)? (5 %)

Expected = 200 × (1, 4, 6, 4, 1)/16 = 12.5, 50, 75, 50, 12.5.

$\chi^2 = 0.5^2/12.5 + 2^2/50 + 1/75 + 6^2/50 + 3.5^2/12.5 = 0.02 + 0.08 + 0.0133 + 0.72 + 0.98 =$ ***1.813***

d.f. = 4 (p given), $\chi^2_{tab} = 9.488 \Rightarrow$ ***accept H0***: binomial fit is acceptable.

9. χ^2 Test for Independence of Attributes

Definition:

Tests whether two attributes (classifications) A and B are independent, using an r × c contingency table.

Formula:

H0: attributes A and B are independent.

Expected frequency E(ij) = (row total i × column total j) / N

$\chi^2 = \Sigma (O - E)^2 / E$, d.f. = (r – 1)(c – 1)

Procedure

- Step 1: Write the observed table with row and column totals. Step 2: compute each expected frequency.
- Step 3: compute χ^2. Step 4: compare with χ^2_{tab} with (r – 1)(c – 1) d.f. Step 5: conclude.

Shortcut for a 2 × 2 table [a b / c d], N = a + b + c + d

Formula:

$\chi^2 = N(ad - bc)^2 / [(a + b)(c + d)(a + c)(b + d)]$, d.f. = 1

Yates' correction (if any cell < 5): $\chi^2 = N(|ad - bc| - N/2)^2 / [(a + b)(c + d)(a + c)(b + d)]$

Example:

Recovery and treatment. Test if recovery is independent of treatment (5 %).

Treatment	Recovered	Not recovered	Total
A	40	10	50
B	30	20	50
Total	70	30	100

- Expected: A: 35, 15; B: 35, 15. $\chi^2 = 25/35 + 25/15 + 25/35 + 25/15 =$ **4.76**
- Check by shortcut: $100(800 - 300)^2/(50 \times 50 \times 70 \times 30) = 25{,}000{,}000/5{,}250{,}000 = 4.76$ ✓
- d.f. = 1, $\chi^2_{tab} = 3.841 \Rightarrow 4.76 > 3.841 \Rightarrow$ ***reject H0***: recovery depends on treatment.
- With Yates: $100(500 - 50)^2/5{,}250{,}000 = 3.857 > 3.841 \Rightarrow$ still reject (barely).

Example:

Gender and preference for three products P, Q, R. Test independence at 5 %.

Gender	P	Q	R	Total
Men	30	40	30	100
Women	20	30	50	100
Total	50	70	80	200

- Expected: Men 25, 35, 40; Women 25, 35, 40.
- $\chi^2 = 25/25 + 25/35 + 100/40 + 25/25 + 25/35 + 100/40 = 1 + 0.714 + 2.5 + 1 + 0.714 + 2.5 =$ ***8.43***
- d.f. = (2 – 1)(3 – 1) = 2, $\chi^2_{tab} = 5.991 \Rightarrow 8.43 > 5.991 \Rightarrow$ ***reject H0***: preference depends on gender.

> ***Note:***
> Independence test: do ***not*** use percentages; and the test says "associated", not "caused by".

10. Practice Problems with Answers

- 1) A sample of 16 has $\bar{X} = 53$ and $s = 4$. Test $\mu = 50$ at 5 %. Ans: $t = 3/(4/4) = 3 > 2.131 \Rightarrow$ reject.
- 2) Two samples $n1 = 7$, $\bar{X}1 = 28$, $s1^2 = 6$; $n2 = 9$, $\bar{X}2 = 24$, $s2^2 = 8$. Test equal means. Ans: pooled $s^2 = (36 + 64)/14 = 7.14$, S.E. $= 2.672 \times 0.504 = 1.347$, $t = 2.97 > 2.145 \Rightarrow$ reject.
- 3) Test H0: σ^2 equal for $s1^2 = 20$ ($n = 13$), $s2^2 = 10$ ($n = 9$). Ans: $F = 2$, d.f. (12, 8), $F_{tab} \approx 3.28 \Rightarrow$ accept.
- 4) 4 categories observed 30, 20, 25, 25 ($N = 100$); equal probabilities? Ans: $E = 25$, $\chi^2 = (25 + 25 + 0 + 0)/25 = 2 < 7.815 \Rightarrow$ accept.
- 5) A 2 × 2 table 20 30 / 40 10: find χ^2. Ans: $N = 100$, $100(200 - 1200)^2/(50 \times 50 \times 60 \times 40) = 100 \times 10^6/6 \times 10^6 = 16.67 \Rightarrow$ reject.
- 6) Why must the F ratio be written with larger variance on top? Ans: so $F \geq 1$ and only upper table values are needed.
- 7) State the properties of the t-distribution and chi-square distribution.
- 8) Distinguish independent t-test and paired t-test.

11. More Worked Examples – t-tests

Example:

Six observations: 2, 4, 6, 8, 10, 12. Test H0: $\mu = 5$ at 5 %.

$\bar{x} = 7$; $\Sigma(x - \bar{x})^2 = 25 + 9 + 1 + 1 + 9 + 25 = 70 \Rightarrow s^2 = 14$, $s = 3.742$.

$t = (7 - 5)/(3.742/\sqrt{6}) = 2/1.5275 = \mathbf{1.31}$; $t_{tab}(5) = 2.571 \Rightarrow$ ***accept H0***.

Example:

Sample C: 22, 24, 26, 28. Sample D: 18, 20, 22, 24, 26. Test equality of means (5 %).

$\bar{x}1 = 25$, $SS1 = 9 + 1 + 1 + 9 = 20$; $\bar{x}2 = 22$, $SS2 = 16 + 4 + 0 + 4 + 16 = 40$.

$s^2 = 60/7 = 8.571$, $s = 2.928$; S.E. $= 2.928\sqrt{(1/4 + 1/5)} = 1.964$.

$t = 3/1.964 = \mathbf{1.53}$; $t_{tab}(7) = 2.365 \Rightarrow$ ***accept H0***.

Example:

Weights (kg) of 6 persons reduce after a diet by d = 3, 5, 2, 4, 6, 4. Is the diet effective? (5 %, one-tailed)

$\bar{d} = 4$; $\Sigma(d - \bar{d})^2 = 1 + 1 + 4 + 0 + 4 + 0 = 10 \Rightarrow s_d^2 = 2$, $s_d = 1.414$.

$t = 4/(1.414/\sqrt{6}) = 4/0.577 = \mathbf{6.93}$; $t_{tab}(5)$ one-tailed $= 2.015 \Rightarrow$ ***reject H0***: diet effective.

Test of significance of a correlation coefficient

Formula:

H0: $\rho = 0$. $t = r\sqrt{(n - 2)} / \sqrt{(1 - r^2)}$, d.f. $= n - 2$

Example:

$r = 0.6$ from $n = 27$ pairs. Is it significant at 5 %?

$t = 0.6 \times 5/\sqrt{(1 - 0.36)} = 3/0.8 = \mathbf{3.75}$; $t_{tab}(25) = 2.060 \Rightarrow$ ***significant***.

Test for a single variance (uses χ^2)

Formula:

H0: $\sigma^2 = \sigma_0^2$. $\chi^2 = (n-1)s^2 / \sigma_0^2$, d.f. = $n-1$

Example:

$n = 10$, $s^2 = 6.4$. Test H0: $\sigma^2 = 4$ at 5 % (right-tailed).

$\chi^2 = 9 \times 6.4/4 = \mathbf{14.4}$; $\chi^2_{tab}(9) = 16.919 \Rightarrow 14.4 < 16.919 \Rightarrow$ ***accept H0***.

12. More Worked Examples – χ^2 Tests

Example:

Genetic ratio 1:2:1 expected. Observed 18, 45, 17 (N = 80). Test at 5 %.

Expected = 20, 40, 20. $\chi^2 = 4/20 + 25/40 + 9/20 = 0.2 + 0.625 + 0.45 =$ ***1.275***

d.f. = 2, $\chi^2_{tab} = 5.991 \Rightarrow$ ***accept H0***.

Example:

Accidents on Mon–Fri were 14, 18, 12, 11, 15 (total 70). Are accidents uniformly spread? (5 %)

Expected = 14 each. $\chi^2 = (0 + 16 + 4 + 9 + 1)/14 = 30/14 =$ ***2.14***

d.f. = 4, $\chi^2_{tab} = 9.488 \Rightarrow$ ***accept H0***: no day effect.

Pooling classes

- If an expected frequency is below 5, combine it with the neighbouring class and recalculate d.f. using the new number of classes.

Example:

Vaccination and attack: test independence at 5 %.

Group	Not attacked	Attacked	Total
Vaccinated	85	15	100
Not vaccinated	60	40	100
Total	145	55	200

- $ad - bc = 85 \times 40 - 15 \times 60 = 3400 - 900 = 2500$.
- $\chi^2 = 200 \times 2500^2 / (100 \times 100 \times 145 \times 55) = 1{,}250{,}000{,}000/79{,}750{,}000 = \mathbf{15.67}$
- d.f. = 1, $\chi^2_{tab} = 3.841 \Rightarrow$ ***reject H0***: vaccination is associated with protection.

Example:

A 3 × 3 table (income level vs. region). Test independence at 5 %.

Region	Low	Medium	High	Total
A	20	30	10	60
B	30	40	30	100
C	10	20	10	40
Total	60	90	50	200

- Expected: A: 18, 27, 15; B: 30, 45, 25; C: 12, 18, 10.
- $\chi^2 = 0.222 + 0.333 + 1.667 + 0 + 0.556 + 1 + 0.333 + 0.222 + 0 = \mathbf{4.33}$
- d.f. = $(3 - 1)(3 - 1) = 4$, $\chi^2_{tab} = 9.488 \Rightarrow 4.33 < 9.488 \Rightarrow$ ***accept H0***: independent.

13. Summary of All Tests

Test	Statistic	d.f.
Large sample mean	$z = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$	–
Large sample proportion	$z = (p - P_0)/\sqrt{(P_0Q_0/n)}$	–

Single mean (small)	$t = (\bar{x} - \mu_0)/(s/\sqrt{n})$	$n - 1$
Two means (small)	$t = (\bar{x}_1 - \bar{x}_2)/[s\sqrt{(1/n_1 + 1/n_2)}]$	$n_1 + n_2 - 2$
Paired	$t = \bar{d}/(s_d/\sqrt{n})$	$n - 1$
Two variances	$F = s_1^2/s_2^2$ (larger/smaller)	$(n_1 - 1, n_2 - 1)$
Single variance	$\chi^2 = (n - 1)s^2/\sigma_0^2$	$n - 1$
Goodness of fit	$\chi^2 = \Sigma(O - E)^2/E$	$k - 1 - m$
Independence	$\chi^2 = \Sigma(O - E)^2/E$	$(r - 1)(c - 1)$

Choosing a test (decision guide)

- Testing means or proportions with $n \geq 30$ → z test (Unit IV).
- Testing means with $n < 30$ and σ unknown → t-test: one sample / two independent samples / paired.
- Comparing the spread of two samples → F-test.
- Counts in categories vs a theory, or two categorical attributes → χ^2 test.

> **Note:**
> Under H0 the statistics t, F and χ^2 are all linked to the normal distribution: $t = z/\sqrt{(\chi^2/\nu)}$, F = ratio of χ^2's.

14. More Practice Problems

- 9) A sample of 10 has $s^2 = 5$ and a second sample of 13 has $s^2 = 3$. Test equal variances. Ans: $F = 5/3 = 1.67$, d.f. (9, 12), $F_{tab} \approx 2.80 \Rightarrow$ accept.
- 10) Heights of 12 men have mean 170 cm, $s = 4$. Test $\mu = 168$ at 5 %. Ans: $t = 2/(4/\sqrt{12}) = 1.73 < 2.201 \Rightarrow$ accept.
- 11) Observed counts 40, 60 and expected 50, 50. χ^2? Ans: $(100 + 100)/50 = 4 > 3.841 \Rightarrow$ reject at 5 % (d.f. 1).
- 12) Why do we use $n - 1$ as d.f. for a single-sample t-test? Ans: one parameter ($\bar{x}$) is estimated from the sample.

- 13) Give 99 % confidence limits for μ from $n = 9$, $\bar{x} = 20$, $s = 3$. Ans: $20 \pm 3.355 \times 1 = (16.65, 23.36)$.
- 14) For χ^2 test of independence in a 4×3 table what are the d.f.? Ans: $3 \times 2 = 6$.

Quick Revision – Unit V

- Single mean: $t = (\bar{x} - \mu_0)/(s/\sqrt{n})$, d.f. $n - 1$.
- Two means: $t = (\bar{x}_1 - \bar{x}_2)/[s\sqrt{(1/n_1 + 1/n_2)}]$, d.f. $n_1 + n_2 - 2$.
- Paired: $t = \bar{d}/(s_d/\sqrt{n})$, d.f. $n - 1$.
- F = larger s^2 / smaller s^2, d.f. $(n_L - 1, n_S - 1)$.
- $\chi^2 = \Sigma(O - E)^2/E$; fit d.f. $= k - 1 - m$; independence d.f. $= (r - 1)(c - 1)$.
- $E(ij) = (\text{row total} \times \text{column total})/N$.